

\$PEACE

DAO Master Report

Confidential internal & partner review.

Engineering and Audit copy

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Draft

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1) Executive Summary

Peace Soldiers requires a governance system that converts verified human impact into durable economic signals, allowing policy to move at the right pace, the treasury to remain disciplined, and the community's voice to be broad yet responsibly bounded. The proposed model is a classed, identity-aware, receipts-first DAO in which every decision maps to a measurable surface—identity, verification, incentives, treasury, and commerce—and every sensitive path is protected by guardrails that fail closed when conditions are not healthy. This design aligns a fixed supply and mechanized scarcity—through a claim-time burn and surplus-only buybacks—with rights linked to the Human Impact Index and with identity-bounded voting. Change is enacted through risk-based quorums and timelocks and governed by a single on-chain Config Registry that makes parameters auditable over time.

Outcome. The resulting system is capture-resistant, budget-disciplined, and operationally observable end to end. Voting power is capped per verified person through the Peace Passport, ensuring that no individual can dominate outcomes. Optional HII-weighting can be activated where appropriate to recognize meaningful, verifiable contributions without undermining fairness. Class-based quorums and minimum timelocks pace change according to risk, ensuring that operational updates move faster than constitutional revisions. Treasury actions, such as buybacks, are conditioned on realized surplus and market hygiene; they are checked against time-weighted pricing, data freshness, and acceptable price deviation, and are automatically blocked if any check fails. Every privileged action emits a receipt that can be verified by machines and reviewed by people, closing the loop between decision, execution, and evidence.

Transparency and verification. Governance, health and core economics are summarized in a signed monthly digest alongside public key performance indicators. These materials are designed to allow independent reviewers to recompute figures and verify the system's state without requiring privileged access. Dashboards surface the same evidence that informs internal operations, reinforcing the principle that trust is earned through reproducible data rather than assurances.

What is distinctive? Unlike typical token-weighted voting systems, the PS DAO hard-codes a per-identity voting cap of **0.1% (10 bps)** and applies classed thresholds—**minimum quorums of 10%, 30%, and 60% with corresponding minimum timelocks of 12, 48, and 96 hours**—so that higher-risk changes require broader participation and more time in public view. A gateway screens proposals for single-subject clarity and parameter bounds before they reach a vote, preventing out-of-range payloads from advancing. The model also leaves room for innovation without compromising safety: AI-assisted reputation signals, fractal sub-DAOs, retroactive impact bounties, and merit-based delegation can be piloted behind timelocks and explicit scopes, promoted only after evidence shows they are safe and beneficial. In short, the PS DAO favors measurable progress over speculation, pairing human-centered legitimacy with disciplined, verifiable execution.

Terminology

Term	Meaning	Defaults / Notes
Identity-bound ed voting	Each verified person can count for at most a fixed share of total voting power, preventing dominance by any single participant.	Cap set at 0.1% (10 bps) per identity; delegation respects the same cap after aggregation.
Classed governance	Changes are grouped by risk; higher-risk classes require broader turnout and longer delays before execution.	Minimum quorums: 10% / 30% / 60%. Minimum timelocks: 12h / 48h / 96h. Community may raise, not bypass.
Mechanized scarcity	Scarcity arises from predictable rules rather than discretion.	Claim-time burn: default 0.75% . Buybacks: surplus-only and executed only when market hygiene checks pass.
Market hygiene checks	Objective safeguards that must be satisfied before treasury actions execute.	Time-weighted average price (TWAP), quote freshness, and price-deviation fences; actions fail closed if any check fails.
Config Registry	The on-chain, authoritative store of parameters and their history.	All changes are proposed, time-locked, executed, and auditable via receipts.
Receipts & Signed Monthly Digest	Evidence artifacts that prove what happened and summarize system health.	Machine-verifiable receipts for privileged actions; a public monthly digest with KPIs designed for third-party recomputation.
Basis points (bps)	A unit for small percentages used throughout governance and treasury policy.	10 bps = 0.1% . Used for vote caps, discounts, and deviation fences.

2) Design Objectives — Grounded in the PS Whitepaper & Tokenomics

The governance design anchors legitimacy in people rather than pure balances, while still rewarding verified contributions. Each participant holds a Peace Passport that confirms their unique identity and grants a bounded share of influence, ensuring that no individual can dominate deliberation. The model permits delegation to trusted stewards, yet the cap remains in effect after aggregation, ensuring that delegated power never exceeds the ceiling. Where appropriate, the Human Impact Index can provide measured uplift to voters who consistently deliver verified, mission-aligned outcomes; however, this uplift is optional, transparent, and never allowed to override the fairness established by the per-identity cap.

Change is paced by risk, so the system can move quickly on routine matters and deliberately on foundational ones. Operational adjustments utilize the lowest quorum and the shortest minimum timelock because they are reversible and low-stakes. Economic or parameterized changes fall in the middle, and constitutional edits—such as altering the identity cap or supply rules—require the broadest turnout and an extended timelock. In practice, this layered approach creates a predictable tempo of decision-making: minor issues are unblocked, while high-impact proposals remain in public view long enough for review, challenge, and preparation.

Treasury actions are automated for discipline and tested for market hygiene before any value is moved. Buybacks are funded solely from realized surpluses rather than discretionary impulses, and they occur only when independent checks—such as time-weighted average price alignment, data freshness, and acceptable deviation—confirm a healthy environment. The system halts rather than improvises if any guard is breached, and every privileged action generates a receipt, allowing auditors and community members to verify not only what was done but also why it was permitted.

A single on-chain Config Registry serves as the project's source of truth for parameters and their history. Rewards and burn rates, HII weights, discount rules in the Store, and the precise thresholds used in market hygiene checks are all maintained in one place with clear diffs and bounded ranges. Because any change must pass through governance and timelock, and because every change is recorded with a receipt, the community can trace the evolution of policy without ambiguity and revert within defined safety rails when needed.

Finally, observability is a first-class objective rather than an afterthought. The DAO publishes signed monthly digests that summarize coverage months, TVDR (as defined in the KPI annex), governance concentration using Gini and Herfindahl measures, de-washed GMV from the Store, buyback activity, and incident counts. These figures are calculated from the same public data streams used to operate the system, allowing third parties to recompute the numbers independently. In this model, confidence comes from reproducible evidence, not from internal assurances.

Terminology (Design Objectives)

Term	Meaning	Default / Practice
Per-identity cap	A ceiling on the voting power any verified person can exercise, protecting the process from dominance.	0.1% (10 bps) per identity; delegation allowed, but the cap applies after aggregation.
Risk-classed change control	Different categories of proposals require different quorums and minimum timelocks according to impact.	Operational, Economic/Parameterized, Constitutional; minimum quorums 10/30/60%; minimum timelocks 12/48/96h.
Mechanized treasury discipline	Rules that govern buybacks and burns based on real surplus and market hygiene, not discretion.	Surplus-only buybacks; hygiene checks on TWAP , freshness , and deviation ; actions fail closed on breach.
Config Registry	The authoritative, on-chain parameter store with diffs and bounded ranges.	All relevant keys (rewards, burns, HII, Store discounts, hygiene guards) change only via governance + timelock.
Observability & KPIs	A reporting layer that allows third parties to recompute system health.	Signed monthly digest; KPIs include coverage months , TVDR , Gini/Herfindahl , de-washed GMV , buyback logs .

Appendix B (Parameter Annex) and Appendix K (Code & Schemas)

2.1. Roles & Entities

1– Security & Treasury Council

2.X Security & Treasury Council (STC)

The Security & Treasury Council (STC) is a small, multi-signature-backed committee responsible for safely executing governance-approved actions on core contracts and treasury programs. The STC is **not** a policy-making body; its mandate is strictly limited to (1) executing decisions that have already passed the required governance process, and (2) triggering emergency shields under well-defined, pre-approved conditions.

2.X.1 Mandate and Scope

The STC operates under the following constraints:

- It MAY execute only those transactions that:
 - originate from an approved PSIP and are present in the official execution queue, or
 - correspond to pre-defined emergency shield actions (e.g., pausing a program, raising a circuit breaker, freezing a compromised address or market) that have been explicitly whitelisted by the DAO.
- It MAY NOT introduce new policy, allocate funds outside of approved ranges, or deploy contracts whose bytecode has not been authorised through the governance process.
- It MUST emit on-chain receipts for every executed action, including metadata linking to the underlying PSIP or shield rule.

The council's operational behaviour is enforced through a combination of:

- multi-signature controls (e.g., 4-of-7 or similar, as defined in the Address & Policy Registry),
- program-level restrictions (allowlists of permissible calls and parameter ranges), and

- auditable off-chain procedures (runbooks, incident logs, SOC-style reporting).

2.X.2 Appointment in Early Phases (Phase 0–1)

In Phase 0–1, when the protocol and governance infrastructure are still being bootstrapped, STC members are appointed by the founding entities (NGO, Foundation and core team). Candidates are selected based on:

- demonstrable experience in security, treasury, or protocol operations;
- verifiable reputation and track record within the ecosystem;
- operational availability (ability to respond within agreed SLOs);
- absence of conflicts of interest beyond what is explicitly disclosed.

During these early phases, the STC is explicitly documented as a **temporary safety layer**. Its composition, mandate and decision logs are made public, and its actions are restricted to a narrow perimeter defined in the Policy & Address Registry.

2.X.3 On-chain Mandates, Terms and Renewal (Phase 1+)

From Phase 1 onwards, STC membership is governed by **on-chain mandates** rather than purely off-chain appointments.

- Each STC seat is associated with:
 - a **fixed mandate duration** (e.g., 12–18 months);
 - a clearly identified **Peace Passport identity** and associated addresses;
 - a maximum number of consecutive mandates (term limits).
- At the end of each mandate, the DAO MUST renew or replace the seat via a C3-class governance proposal (or equivalent high-impact PSIP class).
- Failure to renew a seat within a defined grace period results in the seat being marked as vacant, and the corresponding signer losing their

signing authority for STC operations.

In practice, renewal and rotation follow this pattern:

1. A “STC Mandate Renewal / Rotation” PSIP is submitted, listing the proposed composition for the next term.
2. The PSIP is reviewed by the Governance & Compliance subDAO and the Gateway, ensuring term limits, conflict-of-interest rules and minimum competency requirements are satisfied.
3. Upon approval and after the relevant timelock, the Address & Policy Registry is updated, and the STC multi-signature is rotated accordingly.

2.X.4 Recall and Removal

The DAO retains the power to **recall or remove** one or more STC members at any time via a C3-class proposal, without waiting for the end of the mandate, if any of the following conditions is met:

- material breach of the council mandate;
- consistent failure to meet execution SLOs without acceptable justification;
- evidence of collusion, fraud, or gross negligence;
- regulatory or legal constraints that make continued service incompatible with the DAO’s policies.

A Recall PSIP **MUST**:

- specify which members are to be removed;
- provide a minimal rational and evidence summary (or link to an incident report);
- propose interim or replacement signers, or define a temporary emergency configuration (e.g., reduced quorum, temporary delegation to an audited fallback).

If a recall proposal passes and the timelock expires, the Address & Policy Registry is updated and the multi-signature configuration is changed accordingly. Emergency procedures MAY temporarily downgrade the STC to a smaller quorum if required for continuity, but MUST always be transparently logged and subsequently ratified by the DAO.

2.X.5 Term Limits and Conflict Management

To prevent entrenchment and unhealthy power concentration:

- No individual MAY serve more than N consecutive mandates (e.g., two consecutive terms), as defined in the Parameter Registry.
- After reaching this limit, a cooling-off period (e.g., one mandate cycle) is required before re-appointment.
- STC members MUST disclose material conflicts of interest (e.g., large positions in counterparties, service providers or directly affected protocols).
- In cases of conflict, members MUST abstain from signing relevant transactions; if quorum cannot be achieved, emergency escalation rules apply (e.g., temporary inclusion of pre-approved backup signers).

Together, these rules are designed to ensure that the STC remains a **revocable, accountable and rotating execution layer**, tightly constrained by DAO policy rather than an opaque, permanent multisig.

3) Compliance Map — How the DAO Conforms to PS Tokenomics & Architecture

The DAO maps directly to the commitments defined in the PS Whitepaper and Tokenomics, translating them into enforceable policy and verifiable practice. The fixed supply and mechanized scarcity rules are implemented through bounded keys in the Config Registry and a treasury controller that refuses to act unless market hygiene conditions are met. In other words, supply discipline and value accrual occur when the system is both solvent and healthy, not when sentiment favors it. This approach preserves fiscal credibility while still allowing the community to see, test, and refine the exact thresholds that guard treasury operations.

The principle of identity-bounded voting is enforced at the snapshot and at execution. Each verified participant can contribute up to the per-identity limit. While delegation is encouraged for practical participation, the total counted power after delegation still adheres to the cap. This ensures that influence flows to competent stewards without allowing any person or cluster to grow beyond policy. The system’s gateway also rejects proposals that mix multiple subjects or attempt to exceed parameters beyond their allowed ranges, which keeps voting meaningful and context-specific.

Classed governance formalizes the tempo of change. Operational matters use the lowest quorum and shortest timelock, parameterized changes sit at a higher threshold, and constitutional edits require broad participation and time for discovery and review. By making these minimums explicit and binding, the DAO codifies a professional change-management posture that auditors expect to see: speed where appropriate, scrutiny where required, and clear evidence of both.

A single Config Registry controls the moving parts—rewards schedules and burn rates, HII weights, Store economics, and market hygiene thresholds—so that the DAO’s policy surface is both centralized for clarity and decentralized in control. Every proposal that touches a parameter must produce a diff and a receipt, and any critical action that could shape long-term policy includes a rollback plan. This keeps the system adaptable without letting complexity erode accountability.

Rights and utility linked to the HII are implemented carefully. The model allows HII-aware rights and Store tiers when they improve alignment, while anti-fraud sampling and penalties discourage gaming. Because each privilege is tied to verifiable behavior and each decision leaves a trace, the system can elevate contribution without sacrificing fairness or privacy.

Finally, the DAO’s commitment to transparency is made concrete through monthly signed digests and receipts for each action. These artifacts cover treasury events, governance executions, unlocks, and routine operations. They are intended to be recomputed from publicly available data, which means external reviewers, partners, and regulators can verify the claims without privileged access. This is how the design turns “transparency” from a promise into a measurable fact.

Compliance Matrix (Summary)

PS Canon (Whitepaper & Tokenomics)	DAO Mechanism	Evidence / Reference
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Fixed supply (10B), claim-time burn (0.75%), surplus-only buybacks	Bounded parameters in the Config; treasury controller with TWAP window, quote freshness, and deviation fences; fail-closed execution on breach.	Parameter Annex (App. B); Oracle & Hygiene Notes (App. I); Receipts (App. E); Code & Schemas (App. K).
Identity-bounded voting (0.1% per identity)	Voting snapshot enforces per-ID cap; delegation allowed but capped post-aggregation; delegation auto-expiry.	Governance Design (body text); Parameter Annex (App. B); Receipts (App. E).
Classed governance (10/30/60% with 12/48/96h)	C1/C2/C3 classes enforced by a gateway that blocks out-of-range or multi-subject payloads before voting.	Governance Process (body text); Parameter Annex (App. B); Code & Schemas (App. K).
One Config Registry for rewards/HII/Store	All parameter changes require a diff and a receipt; critical actions include rollback plans.	Config Registry (body text); Receipts (App. E); Monthly Digest (App. H).
HII → rights & utility	Optional HII-weighted mode and Store tiers; anti-fraud sampling and penalties.	Identity & HII (body text); Parameter Annex (App. B); Ops Checklists (App. D).
Transparency & receipts	Monthly signed digests and per-action receipts for buyback_execute, gov_execute, unlock_release, and more.	Governance Communications (App. H); Receipts (App. E).

Terminology (Compliance Map)

Term	Meaning	Auditor Note
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Bound parameter	A key with allowed ranges and validation in the Config Registry.	Prevents out-of-range proposals and enables safe rollback after review.
Fail-closed	If a required check is not met, the system does nothing rather than attempting a risky action.	Expected in treasury controllers and incident response.
Snapshot	The state of voting power captured for a proposal at a specific time.	Enforces the per-identity cap and delegation rules consistently.
Single-subject rule	Each proposal must focus on one change so voters can judge it on its merits.	Reduces ambiguity and improves auditability.
Signed monthly digest	A public report of KPIs and receipts that third parties can recompute.	Strengthens external assurance and reduces reliance on internal statements.

Appendix B (Parameter Annex), Appendix E (Receipt Schemas), Appendix I (Oracle & Data Quality Notes), and Appendix K (Code & Schemas).

4) DAO Architecture

The architecture is designed to maintain clean proposals, fair votes, disciplined treasury actions, and an auditable public record. At the front door sits the **Governance Gateway**, which performs pre-vote checks on every submission. It validates payload structure and schema, verifies that parameters sit within allowed bounds, de-duplicates similar or identical proposals, and applies the appropriate compliance mode. Out-of-range changes are rejected before they reach the ballot, and only proposals that map cleanly to a recognized class are queued. This simple sequencing prevents confusion, reduces spam, and preserves reviewer attention for ideas that are actually eligible for a vote. (Detailed payload formats and hash rules are summarized in prose here and specified in **Appendix K: Code & Schemas.**)

Voting is anchored in the **Peace Passport**, which establishes eligibility and enforces a per-identity cap, ensuring that no single participant can dominate the outcomes. The default voting mode is identity-only, with two optional variations that the community can enable by parameter: a hybrid mode, in which the Human Impact Index provides a modest, transparent uplift for verified contributions, and a quadratic damping mode that reduces the marginal impact of additional voting power. Delegation is available to support practical participation, but delegated influence remains bound by the same post-aggregation cap. Delegations are revocable at any time and, by default, expire automatically on a predictable schedule (for example, every ninety days) unless renewed by both parties. All of these parameters—cap, mode, and expiry—are tracked in the Config Registry and can only be changed through governance.

Proposals are categorized into three classes that reflect the risk associated with the underlying change. *Operational (C1)* items encompass everyday actions, such as manifests and non-privileged operations, and carry the lowest minimum quorum of ten percent and the shortest minimum timelock of twelve hours. *Parameterized (C2)* items adjust bounded policy knobs—such as the TWAP window, data-freshness thresholds, kappa and rho in controller logic, or the coverage floor—and therefore require a higher minimum quorum of thirty percent and a longer minimum timelock of forty-eight hours. *Constitutional (C3)* items define or amend the most sensitive rules—including the per-identity cap, supply discipline, and compliance modes—and demand the broadest participation with a minimum quorum of sixty percent and a minimum timelock of ninety-six hours. Where the constitution requires a supermajority, the higher threshold applies. This classification system aligns the tempo of change with its risk profile and leaves a visible pause before execution, allowing reviewers to raise concerns and operators to prepare.

The **Treasury Controller** converts policy into action, but only when conditions are healthy. Buybacks are funded solely from realized surpluses; they rely on time-weighted prices to reduce manipulation, and they will not execute if market hygiene checks—such as data freshness and price deviation fences—fail. Coverage months act as a runway floor: if reserves cannot sustain the reward program safely, the controller does nothing until the floor is restored. Every permitted action emits a machine-verifiable receipt, allowing the community to retrace what happened, why it was allowed, and how it affected balances. (Oracles and hygiene defaults are explained in prose here; operational details are collected in **Appendix I: Oracle & Data Quality Notes**.)

All policy surfaces are consolidated in a **Config Registry**, the single on-chain source of truth for parameters and their history. Rewards and burn rates, HII weights, Store discount elasticity, market-hygiene thresholds, and distribution caps live behind clear bounds. Any change must pass through governance, leave a diff the community can read, and produce a receipt that shows when it took effect. By placing the configuration in a single auditable location, the DAO avoids “policy drift” and makes rollbacks a controlled and predictable operation.

Finally, the public sees the system through a **Data Plane** designed for recomputation. KPI dashboards present live views of coverage months, TVDR, de-washed GMV, buyback activity, and concentration metrics, while a signed monthly digest package the same facts in a durable format. Because these artifacts are derived from the same streams that drive operations, auditors and partners can verify the numbers independently rather than relying on internal assurances.

Terminology (Architecture)

Term	Meaning	Notes
Governance Gateway	The pre-vote validator that enforces schema correctness, parameter bounds, duplicate suppression, and class routing.	Blocks ineligible or out-of-range proposals before voting.
Peace Passport	The identity credential that establishes eligibility and enforces the per-identity cap.	Delegation allowed; cap applies after aggregation.
C1 / C2 / C3	Operational, Parameterized, and Constitutional proposal classes aligned to risk.	Minimum quorums: 10% / 30% / 60%. Minimum timelocks: 12h / 48h / 96h.
Treasury Controller	Mechanism that executes surplus-only buybacks behind hygiene rails and runway floors.	TWAP alignment, data freshness, and deviation fences; fail-closed on breach.
Config Registry	The single, authoritative parameter store with diffs and receipts.	Keys and bounds changed only via governance.
Public Data Plane	Dashboards and the signed monthly digest that expose system health for recomputation.	Coverage months, TVDR, de-washed GMV, buyback logs, concentration metrics.

Implementation note: Payload schemas, hash methods, controller formulas, and parameter keys are summarized here in prose and specified in **Appendix B (Parameter Annex)**, **Appendix E (Receipt Schemas)**, **Appendix I (Oracle & Data Quality Notes)**, and **Appendix K (Code & Schemas)**.

5) Innovation Modules Piloted Behind Timelocks

Innovation is encouraged but paced, and every new module is bounded by explicit scopes, parameters, and timelocks, allowing the system to learn without taking on hidden risk. The **AI-assisted reputation oracle** utilizes non-transferable “shards” of reputation that decay over time and are only boosted by provable governance contributions. Data arrives through decentralized oracles and can be vetoed by the community if any anomalies are detected. The module is designed for a staged rollout on test networks before any mainnet effect, and it remains a candidate for activation through a parameterized vote.

Fractal sub-DAOs organize work at the edges while preserving the core rules. Each sub-DAO inherits the main policies within defined bounds and connects to the treasury through “membrane” contracts that gate resource flows. Quarterly synchronization votes confirm scope and budgets, and inactivity triggers automatic dissolution to prevent dormant structures from accumulating risk or confusion. This pattern allows local initiative without fragmenting the policy surface.

Liquid-merit delegation strengthens representation without rewarding mere popularity. Delegations are revocable and expire on a regular cadence; stewards can be required to stake an appropriate amount to signal their commitment, and their performance is displayed on public leaderboards that the community can view. When enabled, quadratic dampening reduces the marginal impact of large blocks of delegated votes, further protecting against capture.

Retroactive impact bounties pay for results, not promises. Proposals define measurable outcomes in advance, and payouts are executed only after independent oracles verify that those outcomes have occurred. Multiple oracle routes reduce dispute risk, and vesting smooths distribution to avoid sudden sell pressure. This framework provides contributors with clear targets while keeping treasury risk under control.

Impact circuits dedicate a predictable share of fees to audited social initiatives. Partners publish evidence with optional zero-knowledge proofs when privacy is a concern; the community retains the power to approve, pause, or remove partners through standard governance. By routing funds through an auditable circuit, the DAO links commerce to visible, verifiable impact.

Across all modules, the rule is the same: they remain subordinate to the Config Registry and classed governance. They can help the network move faster and smarter, but they do not gain authority over the core rules, and they never bypass the guardrails that keep the treasury and voters safe.

Terminology (PS Innovation)

Term	Meaning	Notes
Reputation “shards” (SBT)	Non-transferable, decaying attestations of governance contribution.	Issued via oracles; subject to community audit or veto.
Fractal sub-DAO	A smaller governance unit that inherits core rules within bounded autonomy.	“Membranes” gate resource flows and enforce scope.
Liquid-merit delegation	Delegation model with revocation, expiry, and performance visibility.	Optional steward stake; quadratic dampening reduces capture.
Retroactive bounty	Payment released only after verifiable outcomes.	Multi-oracle verification; vesting to reduce dumps.
Impact circuit	A dedicated flow of fees to audited social initiatives.	zk-proofed reports where appropriate; community retains veto.

Pilot status recap: Reputation oracle (pilot candidate via C2), fractal sub-DAOs (C3 charter with C2 budgets), liquid-merit delegation (C2 configuration), retroactive bounties (C2 framework), impact circuits (C2 parameters and partner allowlists).

6) Progressive Decentralization Plan

The path to complete community control is staged so that responsibilities and risks expand together. In **Phase A (Bootstrap)**, the core team authors most proposals to accelerate early setup, while the Council executes only those items that the governance program schedules after a proper vote. Identity caps are active from day one, and frequent audits verify that the gateway, controller, and registry behave exactly as specified. This phase establishes working habits—clear proposals, visible diffs, reliable receipts—before the system scales.

In **Phase B (Shared Control)**, the community elects stewards, the proposer set expands, and economic or parameterized changes move thoroughly on-chain with the same classed protections. Select pilot modules can be enabled through parameterized votes, each with its own guardrails, observable outputs, and sunset conditions if they underperform. The focus of this phase is to distribute authority while maintaining tight accountability and basing it on evidence.

By **Phase C (Mature DAO)**, the most sensitive changes require supermajorities, and external audits are a standing prerequisite for critical upgrades. Emergency powers—while still available for narrowly defined scopes—operate under stricter thresholds and shorter windows, and key authorities rotate on a set schedule to reduce dependency on any single signer or service. At this point, the DAO's processes are not only decentralized but also dependable enough to welcome partners and regulators for thorough review.

Terminology (Decentralization)

Term	Meaning	Notes
Bootstrap Shared / Mature	/ Sequenced phases that expand rights and duties as safeguards harden.	Ensures capability grows in step with accountability.
Supermajority	A threshold higher than a simple majority for critical change.	Applied to constitutional or upgrade actions.
Emergency scope	A narrowly defined set of actions allowed under exceptional conditions.	Time-boxed, logged, and reviewed post-hoc.
Key rotation	Scheduled changeover of upgrade or custody authorities.	Reduces single-point risk; logged via receipts.

Implementation note: Exact thresholds, eligibility rules for stewards, and audit requirements are summarized here and specified in **Appendix B (Parameter Annex)** and **Appendix D (Operational Checklists)**.

7) Success Metrics

The health of governance measures success, the discipline of the treasury, and the quality of real-world impact and commerce. On the **governance** side, we look for dispersed influence rather than concentration. The effective voting power of the ten most prominent participants should remain comfortably below a quarter of the total, and Gini coefficients should trend below 0.60 as participation broadens. Quorum hit-rates by class show whether proposals are attracting sufficient attention, while timelock utilization confirms that changes are spending the intended time in public view. Delegation churn—expiries, revocations, and renewals—offers a window into how actively the community manages representation.

Treasury discipline is tracked through coverage months and hygiene outcomes. Reserves should meet or exceed the policy floor so the system can sustain rewards without stress. Buyback windows should pass hygiene checks almost all the time, with a fail rate trending toward zero. When they do fail, the controller should take no action, and the incident should be visible in dashboards and receipts. Receipt completeness must remain at 100% for all privileged actions so that auditors can reconstruct every step without gaps.

Ultimately, impact and commerce are evaluated based on evidence rather than sentiment. De-washed GMV should grow on a steady path that reflects real transactions rather than noise. HII verification rates should remain high enough to demonstrate that missions are being properly documented. Store discount elasticity should stay within policy bounds so promotions support adoption without eroding the economics that fund buybacks and grants. Together, these measures tell a simple story: a community that can govern itself, a treasury that acts only when safe, and a marketplace that turns real activity into durable value.

Terminology (Metrics)

Term	Meaning	Auditor Note
Top-10 dispersion	Share of voting power held by the ten largest participants.	Target: $\leq 25\%$. Lower is better.
Gini (governance)	Inequality measure of voting power distribution.	Target: < 0.60 . Track trend over time.

Quorum hit-rate	Share of proposals that meet the required turnout for their class.	Indicates engagement and proposal quality.
Timelock utilization	Evidence that executed proposals respected their minimum waiting periods.	Should be 100% compliance.
Delegation churn	Rate of delegation expiries, revocations, and renewals.	Healthy churn implies active oversight.
Coverage months	Months of runway at current settings, using a smoothed outflow estimate.	Must be ≥ policy floor before buybacks.
Hygiene fail-rate	Frequency of windows where TWAP/freshness/deviation checks fail.	Should trend near zero ; fail-closed behavior expected.
Receipt completeness	Presence of receipts for all privileged actions.	Target: 100% completeness.
De-washed GMV	Store volume after removing invalid or suspicious orders.	Correlates with genuine adoption.
Discount elasticity	Sensitivity of demand to Store discount changes.	Must remain within policy bounds.

Cross-references: Formula definitions and calculation windows are summarized here and specified precisely in **Appendix A (KPI Dictionary)** and **Appendix E (Receipt Schemas)**.

Part 2

Core Governance Design & On-Chain Architecture

2.1 Roles & Entities

Citizens (Passport holders). Participation begins with identity. Each verified person may propose (after meeting a clear proposer threshold), vote, and delegate. The per-identity cap ensures no one can dominate outcomes, while delegation allows busy contributors to entrust their voice to stewards without breaching the cap.

Stewards (elected). Stewards raise proposal quality before anything reaches a ballot. They review disclosures, check for conflicts of interest, request clarifications, and assist authors in aligning payloads with the correct class. Their goal is not to decide outcomes but to raise the floor of proposal craftsmanship.

Council (multisig executor). The Council's role is strictly administrative. It co-signs only those transactions that the governance program schedules after a successful vote and completed timelock. It cannot originate actions, modify payloads, or bypass the Config Registry; it serves as a final, transparent handshake between governance decisions and on-chain execution.

Auditors (independent). Auditors publish periodic findings anchored on-chain. They recompute KPIs using public data, sample receipts, and review diffs to verify that the system is meeting policy requirements. Their reports reference immutable artifacts, enabling community members to verify the conclusions independently.

Program Maintainers. Maintainers prepare upgrades under the most stringent governance path. Any change to program code is governed by Constitutional proposals, minimum time locks, and—where required—external audit sign-off before the upgrade authority can rotate.

Terminology — Roles & Entities

Term	Meaning	Notes
Proposer threshold	Minimum criteria to open a vote (e.g., identity age, stake, or support attestations).	Calibrated to deter spam without excluding new talent.

Council	Administrative co-signer that executes only what governance schedules.	No authority to alter payloads or bypass Registry.
Upgrade authority	The key set permitted to deploy program updates.	Rotates only via Constitutional path.

2.2 Proposal Classes, Quorums, and Timelocks (Encoded & Enforceable)

Every proposal is routed into one of three classes that reflect the risk of the change. **Operational (C1)** encompasses routine, non-privileged actions, such as manifests and reporting; it requires a **minimum quorum of 10%** and a **minimum timelock of 12 hours**. **Parameterized (C2)** adjusts bounded policy knobs—such as TWAP windows, oracle freshness, controller sensitivities, or coverage floors—and therefore requires a **minimum quorum of 30%** and a **minimum timelock of 48 hours**. **Constitutional (C3)** governs foundational rules—supply discipline, the per-identity cap, program upgrades, and compliance modes—and demands a **minimum quorum of 60%** and a **minimum timelock of 96 hours**; specific actions may additionally require a supermajority. Out-of-range proposals are rejected at the gateway before voting, which preserves voter attention for in-scope, well-formed questions.

Encoded defaults in the Config Registry make these protections unambiguous and auditable. Quorums are stored as basis-point values (10%, 30%, 60%), and timelocks as minimum seconds matching 12h/48h/96h. Because they are parameters, the community can raise these minimums over time but cannot silently lower or bypass them.

Terminology — Classes & Encodings

Item	Default	Registry Key (summary)
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C1 quorum	10%	<code>gov.quorum.operational_bps</code>	=	1000
C2 quorum	30%	<code>gov.quorum.economic_bps</code>	=	3000
C3 quorum	60%	<code>gov.quorum.critical_bps</code>	=	6000
C1 timelock	≥ 12h	<code>gov.timelock.operational_s</code>	≥	43,200
C2 timelock	≥ 48h	<code>gov.timelock.economic_s</code>	≥	172,800
C3 timelock	≥ 96h	<code>gov.timelock.critical_s</code>	≥	345,600

Note: Full key definitions and bounds are listed in **Appendix B (Parameter Annex)**; schemas in **Appendix K**.

2.3 Voting Mechanics — Identity Before Influence

Eligibility is based on the Peace Passport, which verifies an individual's unique identity. Voting power per identity is capped at **0.1% (10 bps)** of total counted power. For each voter, the system computes raw influence (self plus any inbound delegation) and then counts the **lesser** of that total and the cap. Delegation is encouraged to enhance participation, but it is always revocable and, by default, expires automatically **at the end of each quarter** unless both parties renew. Quorum thresholds depend on class, and within a valid quorum, the outcome is decided by a majority of “For” among the **counted** weight; specific C3 actions may require a supermajority as an additional safety step.

The DAO supports three voting modes. The **identity-only** mode is the default and treats every verified person within the cap equally. A **hybrid (HII-weighted)** mode can be enabled to provide a modest, transparent uplift for sustained, verifiable contribution without breaking fairness. An optional **quadratic damping mode reduces the marginal impact of large voting blocks, further mitigating concentration**. Abstentions register as participation but do not count as approval, which prevents silent acquiescence from being mistaken for consent.

Terminology — Voting

Term	Meaning	Notes
Per-identity cap	Maximum counted weight per person.	0.1% (10 bps) post-delegation.
Counted weight	The lesser of raw influence and the cap.	Ensures fairness even with delegation.
Hybrid mode	Adds modest HII-based uplift to identity votes.	Optional, transparent, bounded.
Quadratic dampening	Reduces marginal impact of large blocks.	Optional anti-capture overlay.

2.4 Governance Gateway — Pre-Vote Guardrail

The gateway is the system's first line of defense. It **hash-locks** payloads to prevent tampering, enforces a **single-subject rule** so voters judge one change at a time, validates the **schema** and parameter ranges, and suppresses duplicates. Proposals that breach Annex-A bounds or attempt to bundle unrelated intents are rejected at this stage. Those that pass are queued into the correct class and exposed for review. The lifecycle is straightforward and traceable: **Draft** → **Gate Check** → **Vote Open** → **Vote Close** → **Timelock** → **Execute** → **Receipt** → **Post-Execution Verification**. This rhythm provides reviewers with a predictable point of intervention, allowing operators sufficient time to prepare for safe execution.

Terminology — Gateway

Term	Meaning	Notes
Hash-lock	Binding a proposal to its content hash.	Prevents post-submission edits.
Single-subject	Exactly one coherent change per proposal.	Improves clarity and auditability.
Bounds	Allowed parameter ranges.	From Annex A in the Registry.

2.5 Config Registry — Single Source of Truth

The Config Registry is the canonical on-chain parameter store with versioning and timelock discipline. Every classed decision that changes policy produces a **machine-readable diff** against the Registry, and every critical change includes a **rollback plan**, allowing the system to return to a safe prior state if needed. Illustrative keys include: the coverage-months threshold that protects runway; a per-window notional cap to limit slippage during buybacks; the per-identity vote cap; class timelock minima; and privacy retention periods for mission evidence. Governance sets parameters and approves automated flows; it **does not** perform ad-hoc transfers.

Promotion rule. Any change to a **Critical** key—such as the vote cap, coverage floor, or buyback notional limits—requires a supermajority and an explicit KPI note explaining the expected impact. This ties adjustments to evidence, making later review straightforward.

Terminology — Registry

Term	Meaning	Notes
Machine-readable diff	Structured record of what changed.	Enables exact review and rollback.
Critical key	Parameter with systemic impact.	Requires supermajority + rollback plan.
Governance scope	Parameters and flows, not direct transfers.	Reduces operational risk.

2.6 Timelocks & Execution

After a successful vote, the proposal is **queued** with an estimated execution time that respects the class timelock. The transparency window invites public comment and targeted spot checks by auditors. When the clock elapses, `dao.execute` routes the exact calldata—anchored to the proposal's content hash—to the target programs. The Council co-signs these transactions as a safeguard against administrative errors. **Receipts** are mandatory for all material actions, including `gov.execute`, `buyback.execute`, `unlock.release`, and emergency events. Each receipt contains canonical fields, allowing third parties to reconstruct what occurred without privileged access.

Terminology — Execution

Term	Meaning	Notes
ETA	Earliest time an approved action may execute.	Must meet the class minimum timelock.
Receipt	Machine-verifiable record of an action.	Standardized fields for audit.
Content hash	Hash of proposal payload.	Binds execution to what was approved.

2.7 Treasury Controller & Buyback Hygiene

The monetary design is conservative and mechanized: a **fixed supply of 10B**, a **claim-time burn of 0.75%**, and **surplus-only buybacks** that execute behind objective guardrails. The controller is **fail-closed**. If price quotes are stale, if time-weighted pricing diverges too far from the spot price, if deviation exceeds the fence, or if coverage months fall below the floor, the controller takes **no action**. Surplus is defined by realized profit, not optics.

Default policy runs buybacks only when three conditions are satisfied: (a) realized profit is present, (b) coverage is $\geq$ **12 months**, and (c) oracle data is fresh. Risk is addressed with **TWAP windows**, **quote-freshness limits**, **per-window notional caps**, and **circuit breakers** for exceptional volatility. All treasury operations generate receipts and are included in the monthly digest, allowing anyone to review what happened and why it was approved.

Terminology — Treasury

Term	Meaning	Notes
Coverage months	Runway at current outflows.	Must meet floor before buybacks.
Hygiene checks	TWAP, freshness, deviation fences.	Any failure → no action.
Fail-closed	Do nothing if checks fail.	Protects reserves from error.

2.8 Identity, Passport & Data Protection

The identity system prioritizes uniqueness over scale, with privacy built in. Each identity event—issuance, renewal, revocation—emits a receipt, creating an auditable trail without exposing unnecessary personal data. The DAO can operate in **Open** or **KYC** mode, based on a Constitutional decision; mode changes follow the highest governance path and are subject to a generous timelock. Practical parameters include a **0.1% cap** by default (policy range **0.05–0.20%**), a **credential TTL of 180 days**, a **Sybil-risk gate** to screen suspicious clusters, a **device cooldown** to slow rapid re-issuance, and an **evidence-retention period of 90 days** with formal purge paths. These controls limit the attack surface while preserving inclusive participation across jurisdictions.

Terminology — Identity

Term	Meaning	Notes
TTL (credential)	Maximum validity period before renewal.	Default 180 days .
Sybil gate	Heuristic threshold for likely duplicates.	Calibrated; policy controlled.
Mode flag	Open vs KYC operating mode.	Constitutional, time-locked.

2.9 Delegation & Stewardship

Delegation is **liquid and revocable**, with **automatic expiry each quarter** to encourage active stewardship. Optional inbound-cap rules prevent any single steward from accumulating excessive delegated power, and modest proposer thresholds or refundable deposits deter spam. The model aligns with the DAO-IDEAS track: **Liquid-Merit Delegation** pairs clear performance dashboards with optional steward staking to signal commitment. Where enabled, a quadratic overlay dampens the marginal impact of very large delegated blocks. All features remain subordinate to the per-identity cap, class quorums, and timelocks, and new configurations are introduced as **C2-gated pilots** with explicit success criteria and sunset conditions.

Terminology — Delegation

Term	Meaning	Notes
Auto-expiry	Delegations lapse on a schedule unless renewed.	Default quarterly .
Inbound cap	Limit on total delegated power to a single steward.	Optional; policy controlled.
Merit dashboards	Public performance views for stewards.	Increases accountability.

2.10 Proposal Lifecycle — Operational SOP

Every proposal follows the same operational cadence. Authors prepare a **disclosure pack** that states the purpose, scope, risks, conflicts of interest, and the intended class. The **gateway** performs bounds, schema, single-subject, and duplicate checks. If eligible, the proposal opens for voting under the appropriate quorum. Once closed, a **timelock** provides a public review window and space for auditor spot checks. The governance program then schedules execution, the **Council** co-signs the hash-derived transactions, and the system **emits receipts**. A brief post-execution verification confirms that the intended state change occurred and that the dashboards accurately reflect the new configuration.

Terminology — Lifecycle

Term	Meaning	Notes
Disclosure pack	Evidence and rationale attached to a proposal.	Improves review quality.
Spot check	Targeted auditor review during timelock.	Prevents rushed errors.
Post-exec verification	Confirmation of effects after execution.	Closes the loop.

2.11 Constitutional Change Process

The constitution defines what requires the strongest protections: supply rules, the per-identity cap, program upgrades, liquidity-lock policy, and compliance mode. These changes use the **C3 path** with a **60% minimum quorum**, a **≥ 96h timelock**, and, where specified, a **supermajority** (for example, two-thirds). A **cooling-off rule** prevents parameter flip-flops within seven days unless the Emergency SOP is invoked. This process establishes a predictable pace of change and the public visibility expected for high-impact decisions.

Terminology — Constitutional Path

Term	Meaning	Notes
Supermajority	Threshold above simple majority for critical change.	Applied to designated C3 actions.
Cooling-off	Delay before re-attempting a recent change.	Default 7 days .
Emergency SOP	Narrow exception path with receipts and expiry.	See §2.13.

2.12 Observability & Public KPIs

From day one, the DAO publishes a **signed monthly digest** and maintains a **public dashboard** that tracks circulating supply, coverage months, TVDR, de-washed GMV, buyback logs, and governance concentration (Herfindahl and Gini). Receipt schemas for governance, treasury, identity, missions, and orders are standardized, allowing anyone to recompute events from public data and reach the same conclusions. This is not marketing; it is an operational commitment to evidence that auditors can test.

Terminology — Observability

Term	Meaning	Notes
Signed digest	Monthly KPI bundle with an on-chain hash.	Anchors reporting. public
Recomputation	Independent re-derivation of KPIs from data.	Core practice. assurance

2.13 Emergency Handling

Two carefully scoped tools address exceptional conditions. **Emergency Pause** is a **time-boxed pause—for example, up to twenty-four hours—triggered by objective conditions, such as an Oracle failure or data corruption**. It is a parameterized action (C2) that emits a receipt and **auto-expires**. **SOP EmergencyStop** is a short pathway for urgent operator response: publish a status page and in-app banner quickly, stabilize the system with a minimal patch, configuration fix, or key rotation, and record the entire sequence for later review. Both tools narrow authority to the specific risk, remain observable, and expire without further action.

Terminology — Emergency

Term	Meaning	Notes
EmergencyPause	Short, scoped halt under clear triggers.	C2; auto-expires; receipt-bound.
EmergencyStop SOP	Operational runbook for urgent fixes.	Status page and banner within hours.

2.14 On-Chain Component Set & Custody Registry

The system consists of a small set of well-scoped programs: a Governance Gateway (serving as a governance proxy and scheduler), a Treasury/Buyback Controller, the Config Registry, a Vesting Escrow, and Liquidity/LP custody, where applicable. Programs are deployed as **upgradeable programs via a buffer and upgrade authority (a modern Solana practice)**, and the **Address & Custody Registry—listing program IDs, upgrade authorities, vaults, and rotation policies—is published post-audit and governed for any modifications**. An example policy would require **4 out of 7 signatures for the Governance Gateway and 90-day key rotation; however, the exact numbers are outlined in the Parameter Annex and the Custody Appendix**.

Terminology — Components & Custody

Term	Meaning	Notes
Upgradeable program	Deployed code with a governed upgrade path.	Buffer + upgrade authority; DAO-controlled.
Custody registry	Canon of addresses, authorities, and policies.	Governance-gated updates only.
Rotation	Scheduled change of sensitive keys.	Reduces single-point risk.

2.15 Attack Surfaces and Controls

The architecture anticipates standard failure modes and ties each to explicit controls. **Governance capture** is mitigated by the per-identity cap, optional HII uplift bounded by policy, quadratic dampening, and class-based quorums with meaningful timelocks. **Market manipulation and thin liquidity** are addressed by TWAP windowing, quote-freshness limits, per-window notional caps, and circuit breakers; if any of these safeguards fail, the controller does not act. **Treasury misuse or optics-driven buybacks** are prevented by the profit-only rule, the coverage-months floor, and receipts that expose every action to monthly review. **Data exposure** risk is reduced through minimal retention, encryption, role-based access, deletion paths, and adherence to standard privacy documents and assessments. The effect is a system that defaults to caution, leaves evidence for later learning, and recovers from stress without improvisation.

Terminology — Risks & Controls

Risk	Primary Control	Secondary Control
Governance capture	Per-identity cap; classed quorums & timelocks	Quadratic dampening; bounded HII uplift
Price manipulation	TWAP & deviation fences	Notional caps; circuit breakers
Treasury misuse	Profit-only rule; coverage floor	Receipts; monthly digest
Data exposure	Minimal retention; RBAC & encryption	Purge paths; DPIA/ROPA

2.16 Specification Clarity

For auditors and engineers, the core decision rules are testable and straightforward. **Vote counting** applies the per-identity cap **after** aggregation, so the counted weight for any person equals the lesser of their raw influence and the cap. The **gateway** rejects proposals that exceed allowed bounds, bundle multiple subjects, or fail schema or hash checks, and it enqueues only those payloads that match a recognized class. **Buyback hygiene** requires a realized profit, a runway at or above the coverage floor, fresh Oracle data, TWAP alignment, and an acceptable deviation; if any of these checks fail, the controller takes no action. Formal pseudocode, IDLs, and alert rules are consolidated in **Appendix K (Code & Schemas)** so that the body text remains readable while the specification remains precise.

Terminology — Spec & Tests

Term	Meaning	Notes
Counted weight	Influence after applying the cap.	Tested per proposal snapshot.
Bounds/Schema checks	Validation of parameters and structure.	Enforced at the gateway.
Hygiene pass	All market checks satisfied.	Required before treasury action.

Cross-references: Detailed parameter bounds in **Appendix B**, receipt formats in **Appendix E**, oracle practices in **Appendix I**, and full pseudocode/IDLs in **Appendix K**.

2) Timelocks

3.X Timelocks, Execution SLOs and Permissionless Execution

Timelocks are a core element of the Peace Soldiers governance architecture. They provide a predictable buffer between “**decision**” and “**effect**”, enabling stakeholders to review, react and, if necessary, intervene before a change is applied on-chain. To ensure timelocks remain a safety mechanism rather than an operational bottleneck, Peace Soldiers combines them with clear **execution SLOs (Service-Level Objectives)** and, where feasible, **permissionless execution**.

3.X.1 Timelock Classes and Durations

Governance actions are grouped into three primary classes, each associated with different timelock durations and risk profiles:

- **C1 – Urgent / Security-Critical Changes**
 - Examples: pausing a compromised program; freezing a clearly malicious address; raising a critical shield; switching traffic away from a failing provider.
 - Timelock: short (e.g., 12–24 hours), to allow rapid response while still offering a brief review window.
- **C2 – Standard Policy & Parameter Changes**
 - Examples: adjusting certain treasury allocation bands within predefined ranges; tuning incentive parameters; updating non-critical program configs.
 - Timelock: medium (e.g., 48–72 hours), providing sufficient time for community review and discussion.
- **C3 – High-Impact / Constitutional Changes**
 - Examples: altering core governance rules; changing emission curves; modifying the core treasury strategy framework; changing key program addresses.

- Timelock: long (e.g., 96 hours or more), to reflect higher stakes and the need for deeper scrutiny.

Exact durations for C1/C2/C3 are parameterised in the Governance Parameter Registry and can be updated via C3-class proposals.

3.X.2 Execution SLOs (Service-Level Objectives)

A timelock by itself does not guarantee reliable execution. To avoid situations where approved proposals remain dormant in queues, the DAO defines explicit execution SLOs for each class:

- For **C1** proposals, the target is that execution occurs within *E1* hours after timelock expiry (e.g., within 6 hours), except where a shield rule or veto is triggered.
- For **C2** proposals, the target is execution within *E2* hours after timelock expiry (e.g., within 24 hours).
- For **C3** proposals, the target is execution within *E3* hours (e.g., within 48 hours).

The STC and relevant subDAOs are accountable for meeting these SLOs. Persistent breach of execution SLOs, without acceptable justification, constitutes grounds for council review or recall. Execution SLO performance is tracked in the DAO's operational KPIs and reported in periodic governance transparency reports.

3.X.3 Permissionless Execution After Timelock

Wherever it is technically and economically feasible, Peace Soldiers adopts a **permissionless execution** pattern:

- Once a proposal has:
 - passed the required governance vote;
 - completed its timelock period; and
 - not been cancelled or vetoed by an authorised shield, it becomes *eligible for permissionless execution*.

- In this state, **any address** may call the `execute()` function on the queued transaction.
- The STC retains only the right to **cancel** or **veto** during the timelock window, not to delay or block execution after the window has expired.
- If the transaction includes operations which are only partially executable (e.g., because pre-conditions have changed), the transaction **MUST** either fail atomically or emit detailed receipts indicating which sub-operations were skipped and why.

This pattern reduces the risk that governance outcomes become hostage to the availability or willingness of a small set of signers, while still retaining safety controls aligned with shield rules and emergency policies.

3.X.4 Fallbacks, Vetoes and Shield Rules

Timelocks are integrated with the DAO's adaptive shields and veto mechanisms:

- **Before timelock expiry:**
 - Authorised entities (e.g., the Risk & Security subDAO, STC under shield rules) **MAY** cancel or veto a queued transaction if specific, pre-approved conditions are met (for example, discovery of a critical bug, a major security incident, or regulatory prohibition).
 - Any such action **MUST** emit a public receipt with a reference to the underlying incident or rule.
- **After timelock expiry:**
 - Cancellation or veto is only allowed under the narrow conditions defined in shield policies (e.g., newly discovered catastrophic risk).
 - In normal circumstances, veto powers lapse and the transaction **SHOULD** be executed within the SLO window, by anyone.

If a veto is triggered after timelock expiry, a follow-up governance proposal (e.g., a C2 or C3 PSIP, depending on impact) **MUST** be submitted that:

- documents the rationale;
- proposes remediation or alternative actions; and
- revalidates confidence in the governance and execution pipeline.

3.X.5 Monitoring and Reporting

To ensure the timelock and execution system behaves as intended, the DAO commits to:

- tracking for each proposal: creation time, vote outcome, timelock start/end, execution time, executor address and result;
- publishing regular summary metrics, including:
 - median and distribution of execution delay vs. SLO;
 - number of cancelled or vetoed proposals and reasons;
 - distribution of executors (how concentrated or decentralised actual execution is);
- reviewing these metrics periodically and updating parameters (timelock lengths, SLOs, shield rules) via governance when systemic issues are identified.

In combination, timelocks, SLOs, permissionless execution and shield rules create a governance pipeline that is both **safe** and **operationally reliable**: stakeholders have time to react, but decisions that have passed are not left in limbo.

2.18 Compliance Reaffirmation — Tokenomics ↔ Governance

The monetary logic of Peace Soldiers is not a set of discretionary practices; it is a governed surface with explicit, enforceable bounds. Claim-time burn, surplus-only buybacks, and the hygiene rails that protect buyback execution—coverage floor, TWAP alignment, and quote freshness—are each parameterized in the Config Registry and can only change through the appropriate class of proposal. Routine tuning of bounded “knobs” proceeds via Parameterized (C2) decisions with clear ranges and receipts; any attempt to alter systemic levers, such as the per-identity vote cap or supply discipline, is treated as a Constitutional (C3) action that requires a higher quorum, longer timelock, and (where specified) supermajority approval. In parallel, identity-bounded voting and classed change control are encoded as defaults in the Registry and promoted to constitutional status, ensuring that future alterations follow the slowest, most transparent path. The net effect is a closed loop: tokenomics define what must be guarded, governance defines how change is allowed to occur, and the Registry records exactly what changed and why—so auditors can reconcile policy with on-chain reality without guesswork.

Terminology — Compliance Reaffirmation

Term	Meaning	Notes
C2 / C3 governance	Parameterized vs. Constitutional change paths.	C2 for bounded tuning; C3 for systemic rules.
Hygiene rails	Coverage, TWAP, and freshness checks gating buybacks.	Any failure → no action (fail-closed).
Constitutional promotion	Elevating defaults to the strongest governance path.	Applies to identity cap, upgrades, supply discipline.

Part 2 — Deliverables Recap

This part provides a comprehensive, auditable blueprint for core governance and its on-chain manifestation. It defines who can act and how (Citizens, Stewards, Council, Auditors, Maintainers), frames proposals through classed quorums and timelocks, and encodes voting such that identity precedes influence while still allowing practical delegation under a hard cap. It explains the Governance Gateway, which filters malformed or out-of-range payloads before any vote, and positions the Config Registry

as the single source of truth, whose diffs and receipts make policy evolution traceable. Execution is queued and time-locked with content-hash binding, receipts are mandatory for all material actions, and the Treasury Controller remains fail-closed behind hygiene rails and a coverage floor. Identity and retention follow privacy-by-construction, with receipts for each credential event. Emergency procedures are narrowly scoped and time-boxed, and the custody registry clarifies which keys control what and how rotations occur. Finally, it integrates innovation modules—steward narratives, reputation oracles, and fractal sub-DAOs—only as C2-gated pilots fully subordinate to Registry bounds and classed governance, so experimentation never outruns safety.

Quick recap (for orientation): roles and authorities; proposal classes and encoded quorums/timelocks; vote math with the 0.1% per-identity cap; gateway schema/bounds checks; Registry keys and diffs; timelock/execute with receipts; treasury hygiene and runway floors; identity issuance/retention; emergency SOPs; custody registry; and pilot modules gated behind C2 with sunset rules. Cross-references: **Appendix B** (Parameter Annex), **Appendix E** (Receipt Schemas), **Appendix I** (Oracle & Data Quality Notes), **Appendix K** (Code & Schemas).

Terminology — Part 2 Recap

Term	Meaning	Notes
Content-hash binding	Execution is tied to the exact payload that passed.	Prevents post-vote modifications.
Receipt completeness	Presence of receipts for all privileged actions.	Target: 100% for auditability.
Custody registry	Canon of program IDs, vaults, authorities, and rotation rules.	Governance-gated updates only.

Part 3 — Comparative Research & Audited Lessons

3.1 Survey of Dominant DAO Designs

MakerDAO (Endgame, MIPs, and RWA Program)

How it works. MakerDAO formalized change through Maker Improvement Proposals (MIPs) and a written constitution that clarifies roles and decision paths. Delegates and executive votes administer parameter changes, while the project expanded its income base with real-world asset vaults—including programs that invested stable reserves into short-duration government securities via an off-chain legal wrapper. A well-maintained governance portal centralizes proposals, votes, and the status of execution.

Strengths. The MIP framework and the separation of signal vs. executive stages create a clear paper trail and cadence. The RWA pipeline diversified revenue and stabilized the stablecoin's economics during periods of on-chain yield compression. Process maturity—risk memos, forums, and auditable artifacts—sets a benchmark for public record-keeping.

Weaknesses. Complexity and cognitive load are significant. The breadth of scopes and committees increases political coordination overhead, which can slow critical decisions.

Takeaways for PS. Keep the **Config Registry** minimal yet explicit: name the parameters, define owners, publish diffs, and attach receipts. Avoid sprawling process layers; adopt a compact **PSIP** standard with tight SLAs and classed quorums/timelocks encoded at the contract layer. PS should adopt the documentation discipline of Maker's risk memos but preserve leaner routes to bounded parameter tuning. *(Specific RWA structures and timelines are summarized here and detailed in **Annex C-1** for auditors.)*

Aragon OSx (Modular EVM Framework) and the Treasury Lesson

How it works. Aragon OSx composes governance through installable plugins—governance, membership, upgradeability—rather than a monolithic contract. This modularity lowers the cost of adding or removing features as a DAO's needs evolve.

Turning point. In 2023–2024, the Aragon Association dissolved and returned a large share of its non-native treasury to token holders under public pressure. The episode

highlights the delicate interface between on-chain governance, off-chain legal entities, and treasury control.

Takeaways for PS. Favor **on-chain, policy-gated treasury flows** with multi-sig enforcement and ex-ante constraints; redemptions or returns, if ever required, should be **policy-driven rather than ad hoc**. Emulate modularity by using **program-upgrade gates and feature flags** on Solana so PS can pilot add-ons (e.g., AI screening, reputation gating) without touching the core. *(Event specifics and redemption mechanics are summarized here and detailed in **Annex C-2**.)*

DAOstack (Holographic Consensus)

How it works. Holographic Consensus attempts to scale governance attention by letting curators stake on proposals to “boost” them into wider consideration. The design combines a signaling token with limited voter bandwidth to filter noise.

Strengths. Attention-scaling and curation can surface high-signal items and reduce voter fatigue.

Weaknesses. Reliance on a side token and curator game theory adds complexity, and ecosystem traction has waned as competing models mature. Academic work documents both the benefits and trade-offs.

Takeaways for PS. Retain the **attention-filtering idea** but avoid dual-token complexity: use **PS-native reputation**, bounded by identity caps, and simple stake-or-attestation signals to prioritize reviews without creating a new asset. *(Comparative notes and evaluation references are summarized here and expanded in **Annex C-3**.)*

Compound / Governor Bravo (and the OpenZeppelin Governor Lineage)

How it works. The Governor pattern popularized a clear lifecycle—proposal, voting with delegation, quorum checks, timelock, and queued execution—backed by widely used implementations.

Incident to learn from. A production upgrade once produced excess rewards, requiring follow-up governance actions to repair the state. The lesson is procedural: stage upgrades, simulate economically, and preserve kill-switches.

Takeaways for PS. Adopt the **Governor-style lifecycle** but enforce PS’s **0.1% per-identity cap** and **classed quorum/timelock** at the program level. Require

pre-deployment simulations, canary executions on testnet, and an emergency pause scoped to governance scheduling only—never as a backdoor to move treasury funds outside of the time-locked path. *(Incident specifics and remediation sequence are summarized here and detailed in **Annex C-4.**)*

Optimism Collective (Bicameral Governance & RetroPGF)

How it works. Optimism separates responsibilities across two houses: a Token House that votes based on token weight and a Citizens’ House that allocates Retroactive Public Goods Funding using a one-person-one-vote system. The model strikes a balance between plutocratic efficiency and human-centric legitimacy.

Strengths. Role clarity and a production-grade RetroPGF pipeline have made significant public-goods allocations possible without collapsing into either a pure token rule or a pure identity rule.

Weaknesses. Identity issuance and sybil-resistance are non-trivial, and the bicameral structure adds operational complexity.

Takeaways for PS. PS already centers **identity-bounded voting**; it can further incorporate a **Citizens Panel** for mission impact decisions, mirroring RetroPGF in spirit but binding it to **HII-verified outcomes**. This keeps public goods funding accountable to evidence rather than narrative. *(Program parameters and operator notes summarized here; fuller references in **Annex C-5.**)*

Gitcoin DAO (Quadratic Funding, Passport)

How it works. Gitcoin’s Grants Stack (on the Allo protocol) supports grant rounds with **quadratic funding**, while **Passport/Human Passport** provides sybil-resistant attestations to protect matching pools.

Strengths. Mature public-goods operations and well-documented anti-sybil approaches.

Weaknesses. Identity frictions are real: the more robust the checks, the more participation hurdles; the design must trade off privacy, inclusivity, and attack resistance.

Takeaways for PS. Integrate **sybil-resistant attestations** for authors and voters, layered under PS’s **0.1% identity cap** and optional **quadratic dampening**. Borrow the operational playbooks—transparency in round design and clear rejection reasons—without importing complexity that does not match PS’s mission footprint. *(Implementation references summarized; details in **Annex C-6.**)*

Jupiter (Pragmatic Phasing of Community Governance)

State of play. Jupiter signaled that formal DAO voting would be paused until a later date while the team focused on shipping cores and stabilizing operations. In practice, a pause does not equal absence of governance; it formalizes **milestones** and **preconditions** before opening the gates.

Takeaways for PS. If PS ever phases out governance, write the schedule and conditions into the Config Registry—including KPIs and audit gates—and publish them in advance. That way, a temporary pause becomes a documented runway to a defined opening rather than an open-ended delay. (*Public communications summarized; chronology in **Annex C-7.***)

Terminology — Comparative Survey

Term	Meaning
MIP (Maker Improvement Proposal)	Maker's formal process for proposing and ratifying changes.
Executive vote	Maker's binding on-chain action following signal and review.
RWA (Real-World Assets)	Off-chain financial instruments onboarded via legal wrappers and on-chain vaults.
Holographic Consensus	Attention-scaling mechanism using curator stakes to "boost" proposals.
Governor / Timelock	Widely used pattern for proposal → vote → timelock → execution.
RetroPGF	Retroactive Public Goods Funding paid after verifiable impact.
Passport Passport / Human	Sybil-resistant attestations for participants in funding rounds.

Numerical specifics, timelines, and source links are summarized in the body and expanded in **Annex C (Comparative Dossiers)**.

3.2 Case Studies & Failure Analyses

The DAO (2016): Reentrancy and Governance Immaturity.

The original DAO's collapse was a software failure amplified by governance inexperience. A reentrancy vulnerability lets an attacker drain funds by repeatedly calling a withdrawal function before the balances are updated. The episode highlighted how brittle complex systems can become when they lack audited patterns, emergency stoppages, and a conservative rollout plan. It also reminded projects that communications must be precise: investors and users should never be promised rights or returns that the system cannot legally or technically guarantee. For PS, the enduring lesson is procedural discipline. Use established patterns, require independent security reviews, introduce **pause/timelock** layers that default to safety, and stage features through testnets and limited-scope pilots before opening participation. *(Technical chronology summarized here; specifics and timelines are provided in **Annex C-8**.)*

Compound (2021): Governance Upgrade Bug.

A governance upgrade introduced a miscalculation in reward accrual that distributed an estimated \$80M in COMP. The team and community recovered through successive proposals that limited claims and patched logic; however, the event demonstrated that a well-loved pattern (Governor + Timelock) still fails without end-to-end simulation. The PS corollary is unambiguous: before any mainnet change, run **shadow-fork tests**, economic simulations, and canary executions. Maintain a **post-vote delay** so production users can scrutinize changes, and add **circuit breakers** to pause abnormal reward flows automatically if telemetry diverges from expected ranges. *(Incident details and remediation sequence in **Annex C-4**.)*

Parity Multisig Freeze (2017).

A flaw in a shared library allowed it to self-destruct, bricking multisig wallets and freezing roughly \$150–\$160 million of ETH. The root cause was not a single “bug” so much as a fragile dependency surface: many wallets shared a library with dangerous privilege and no safe upgrade path. PS should therefore minimize shared, non-upgradeable libraries; keep upgrade authority distributed and well-documented; and schedule continuous audits for the **dependency supply chain**, not just for application code. *(Technical notes and mitigation patterns in **Annex C-9**.)*

Curve/Vyper Exploits (2023).

Compiler regressions affected a family of pools, illustrating how toolchain issues can ripple across multiple deployed systems. Even protocols with rigorous internal checks were impacted by an upstream vulnerability. PS's response model treats the toolchain as a first-class risk: pinning compiler versions, maintaining supply-chain audit routines, and keeping a **Treasury Risk Playbook** for correlated stress—covering liquidity buffers,

temporary parameter tightening, and incident communications. *(Affected versions and containment strategies summarized; references in **Annex C-10.**)*

Aragon Treasury Return (2023–2024).

Under sustained governance pressure, the Aragon Association dissolved and returned a significant portion of its non-native treasury to token holders. The decision, shaped by legal wrappers and off-chain governance, shows how vague custody mandates can catalyze extreme outcomes. PS should codify **who can redeem what and when**, express constraints **ex-ante** in the Config Registry, and ensure that any redemption or distribution event proceeds via publicly reviewed policy rather than ad-hoc discretion. *(Treasury chronology and wrapper analysis in **Annex C-2.**)*

Terminology — Case Studies

Term	Meaning	Auditor Note
Reentrancy	A contract call pattern that allows state-changing logic to be re-entered before completion.	Prevented by mutex patterns and audited templates.
Shadow-fork	A private fork of mainnet used for realistic testing.	Mandatory before upgrades.
Supply-chain audit	Review of compilers, libraries, and build tools.	Includes version pinning and regression checks.
Playbook	Pre-committed response steps for incidents.	Reduces improvisation risk.

3.3 What PS Should Adopt, Adapt, and Avoid

Lifecycle. PS should adopt the **Governor-style lifecycle**—proposal, vote, timelock, execution—because it is transparent, auditable, and familiar to both users and auditors. PS adapts this baseline by **encoding classed quorums and timelocks** directly in program logic for safety-critical parameters so they cannot be bypassed. PS should avoid any “instant executive” paths that allow privileged changes without a public waiting period.

Voting power. Delegation, receipts, and transparent tallies belong in PS, but influence must remain **identity-bounded** at **0.1% per person**, with a challenge and appeal process for suspected fraudulent identities. PS should avoid pure token-weight voting without identity limits, as it invites concentration and reputational risk.

Attention scaling. Queue-based curation raises proposal quality. PS adapts the **DAOstack “boosting” idea** by grounding it in **PS-native reputation** and identity caps instead of a side token, avoiding the complexity of curator markets. Tokenized curation assets should be avoided.

Public goods. Optimism’s bicameral instinct is correct: human-centered checks matter. PS adapts this with a **Citizens Panel** focused on HII-verified missions and **RetroPGF-style** allocations, which are paid only after a measured impact. A single house charged with every decision should be avoided.

Treasury and commerce loop. PS should adopt timelocked, module-gated execution; adapt with **surplus-only buybacks** protected by registry guardrails and role constraints; and avoid discretionary, off-chain treasury hand-offs. In commerce, PS ties **Peace Store fees** → **treasury** → **scheduled buybacks** per policy, and avoids hype-led emissions disconnected from verified activity.

Rollout. PS adopts **shadow forks and canary deployments**, **adapts with narrowly scoped kill switches for governance flows**, and **avoids hot patching on the mainnet**.

Terminology — Adopt/Adapt/Avoid

Theme	Adopt	Adapt	Avoid
Lifecycle	Governor + Timelock	Encode classed minima on-chain	Instant executives
Voting	Delegation + receipts	0.1% cap + fraud challenge	Pure token weight
Attention	Curation queues	Reputation-based boosting	Side-token curators
Public goods	Bicameral checks	HII-tied Citizens Panel	One-house for all
Treasury	Timelocked modules	Surplus-only, guarded	Off-chain discretion
Rollout	Shadow-forks	Kill-switches for gov flows	Hot-patches

3.4 Off-Chain Governance Infrastructure

Healthy on-chain mechanics require an equally disciplined off-chain substrate. PS maintains a **public forum** with an RFC and **PSIP** template so proposals arrive with consistent disclosures, risk rationales, and clear class intent. A **delegate registry** records statements of purpose, voting history, and conflicts to help citizens choose stewards and to support post-hoc review. **Research and risk committees** operate under **bounded mandates** and **automatic renewal votes**, which keeps expertise in the loop without creating permanent councils. When a legal wrapper is needed, it serves narrow, **non-discretionary** functions—such as acting as registrar of record or KYC escrow for vendors—but it never overrides the chain or controls treasury flows; PS encodes that boundary upfront to avoid Aragon-style ambiguity. (*Forum templates and registry fields summarized here; complete formats in **Annex C-11**.*)

Terminology — Off-Chain Infra

Term	Meaning	Notes
PSIP	Peace Soldiers Improvement Proposal template.	Standardizes disclosures.
Delegate registry	Public record of delegates' positions and history.	Enables informed delegation.
Bounded mandate	Committee scope with explicit renewal.	Prevents permanent councils.
Registrar role	Legal entity for signatures/compliance only.	No treasury discretion.

3.5 Controls PS Should Encode On-Chain

PS hard-codes **parameter safety rails** in the Config Registry. Each key names its owner scope, quorum class, **maximum change per proposal**, and timelocks; if a change would raise risk, the required quorum and delay increase accordingly. Voting remains **identity-bounded** at 0.1% per person, enforced at snapshot and execution; when fraud flags appear, a **challenge route** temporarily suspends the identity while an appeal process runs on evidence. The **Governance Gateway** improves proposal quality with two filters: a **reputation-based attention boost** rooted in PS-native signals (mission

completions, delivery proof, forum reputation), and **AI-assisted triage** that flags duplicates, missing disclosures, or non-compliant payloads before a vote opens.

Treasury rules are constrained by **invariants**: buybacks are **surplus-only** on a fixed cadence; **programmatic halts** trigger when revenue drops or volatility spikes beyond policy; and payouts proceed in **two stages** (approve → fund stream) with **clawback windows** for non-delivery. An **emergency pause** can halt governance scheduling but cannot seize funds that are already time-locked. The **commerce flywheel** routes Peace Store fees directly to the treasury module and then to buybacks, as per policy. Finally, **HII-verified missions** feed a PS-specific allocation channel—similar in spirit to RetroPGF—so the system rewards measurable impact rather than promises. (Key definitions and bounds in **Appendix B**; gating logic in **Appendix K**.)

Terminology — Encoded Controls

Control	Meaning	Auditor Note
Max delta	Largest allowed param change per proposal.	Prevents step-changes without broad buy-in.
Challenge route	Temporary suspension + appeal flow for identity fraud.	Evidence-based, time-boxed.
Two-stage payout	Approve first, stream later with clawback.	Reduces non-delivery risk.
Flywheel	Store fees → treasury → buybacks.	Policy-driven, receipt-logged.

3.6 Implementation References

PS draws from several proven designs while maintaining a compact core. **Maker’s Endgame and Constitution** demonstrate how to formalize guardrails and publish living records; PS encodes its version **on-chain** via the Config Registry and receipts rather than sprawling documents. **Aragon OSx** demonstrates the value of modularity; PS mirrors this with program-level feature flags and governed upgrades, rather than monolithic redeploys. **The Compound’s Governor and Timelock remain the operational backbone**, with PS extending it by enforcing identity caps and class minima in code, and by mandating **staging and simulation** before release. **Optimism’s bicameral checks** inform PS’s **Citizens Panel** for public-goods allocation, while **Bitcoin’s Passport** experience guides PS’s **sybil-resistant attestations** and

privacy-sensitive participation. Finally, **Jupiter's phased posture** reinforces PS's commitment to **pre-committed milestones** when deferring votes. (*Primary mappings and rationale tables in **Annex C-12**.*)

Terminology — References

Term	Meaning	Notes
Feature flag	Governed switch to enable/disable a module.	Avoids core redeploys.
Staging	Pre-production testing in a realistic environment.	Shadow-forks + canaries.

3.7 Checklist Distilled from Failures

The common thread across historical incidents is not just bad actors alone, but also **missing guardrails**. PS encodes those guardrails by policy and by program. Every governance payload undergoes shadow-fork tests, economic simulations, and human-in-the-loop dry runs before being deployed to the mainnet. Critical changes are subject to a **minimum T+48h** delay even after approval, while emergencies are **narrowly scoped and receipt-bound**. Toolchains are treated as risk surfaces: compilers and libraries are **pinned**, shared libraries are minimized, and the **supply chain** receives scheduled audits. Legal wrappers never outrank code: they function as **registrars**, not treasury owners. Identity legitimacy is maintained through **proofs plus caps**, supported by an **appeals process** with documented evidentiary standards. These practices do not slow progress; they prevent costly rollbacks and reputational damage.

Compact checklist recap

- **Staging & Simulation:** shadow-forks, sims, canaries before mainnet.
- **Timelocks Everywhere:** Minimum T+48 hours for criticals; visible emergency windows.
- **Dependency Hygiene:** version pinning; shared-library reduction; supply-chain audits.
- **Legal Non-Discretion:** registrars only; no off-chain treasury override.

- **Identity & Caps:** proofs + 0.1% cap; challenge and appeal flow.

Terminology — Checklist

Term	Meaning	Auditor Note
T+48h	Earliest execution 48h after approval.	Extends review and detection time.
Canary	Limited-scope production test.	Low blast radius.

3.8 Direct Alignment with PS Whitepaper & Tokenomics

The governance layer implements the whitepaper’s human-first legitimacy with a **0.1% per-identity cap**, **delegation under the cap**, and **classed quorums/timelocks** that make sensitive changes slow and visible. The value loop described in tokenomics is enforced by code and data: **Peace Store fees** accrue to the treasury; buybacks are **surplus-only** and gated behind **TWAP/freshness/deviation** checks; and a **0.75% claim-time burn** ties scarcity to verified human activity. Impact flows do not rely on rhetoric; **HII-verified missions** influence a **Citizens Panel** allocation track that pays **retroactively** for measured outcomes. Each of these surfaces is parameterized in the Config Registry with machine-readable diffs and receipts, allowing auditors to reconcile policy with execution line by line. (*Cross-references: **Appendix A KPIs**; **Appendix B Parameter Annex**; **Appendix E Receipts**.*)

Terminology — Alignment

Term	Meaning	Notes
Claim-time (0.75%)	burn Supply reduction applied at reward claim.	Mechanized scarcity.
Surplus-only buybacks	Purchases funded by realized profit.	Hygiene-gated; fail-closed.
Citizens Panel	Identity-based allocation track for impact.	Retroactive, HII-verified.

3.9 Summary

PS's governance blueprint borrows the best-tested mechanics in the field and binds them to a human-centered legitimacy model. From the **Governor lifecycle** and **timelocks** to **identity caps** and **classed quorums**, every safeguard that history recommends is encoded where it matters—in parameters, in programs, and in receipts. Where others rely on trust, PS relies on **evidence**: monthly digests, recomputable KPIs, and public receipts make assurances verifiable. Innovation is welcomed, but only through **time-locked pilots** and **feature flags** that never outrun safety. The result is a DAO that reads well to non-experts, satisfies auditors, and, most importantly, aligns real human impact with durable on-chain value.

Terminology — Summary

Term	Meaning	Auditor Note
Classed governance	Risk-tiered quorums and timelocks.	Encoded minima; raiseable only.
Evidence-first	Receipts + recomputable KPIs.	Trust through verification.
Feature-flagged pilots	Timelocked trials under bounds.	Subordinate to Registry and caps.

4) Architecture & Smart-Contract Blueprint

4.1 Target Stack & Design Posture

The core architecture of the Peace Soldiers (PS) DAO is designed to leverage Solana's high-performance infrastructure, using **Rust** and **Anchor** for program development. The governance model follows a clear, auditable lifecycle: **proposal** → **vote** → **timelock** → **execution**, with all critical parameters encoded at the contract layer for transparency and accountability.

The PS DAO consists of several core programs and components:

- **DAO**: The governance gateway that handles proposals, voting, timelocks, and execution.
- **Config**: The on-chain parameter registry that governs all mutable system settings.
- **Treasury**: The surplus-only buyback controller, which manages buybacks and burn operations with strict hygiene checks.
- **Store**: The Peace Store commerce plane, integrating with the treasury and reward systems.
- **Vesting**: The vesting and escrow program for token allocation and locked funds.
- **Indexers**: Off-chain services that monitor and compute key performance indicators (KPIs) for transparent reporting.

The DAO integrates with **Pyth** as the primary oracle for market data, complemented by **Switchboard** for redundancy. The oracles are equipped with strict time-weighted average price (TWAP) checks, quote freshness validation, and deviation limits to ensure reliable data before treasury actions are taken.

Upgrades are handled via Solana's **BPF-Upgradeable Loader**, with the upgrade authority held by a **DAO-controlled key** (e.g., multisig or realm) and rotated only through **C3 Constitutional proposals** to maintain transparency and security.

4.2 Program Suite Overview (Solana Accounts, PDAs, Instructions)

A) Governance Program — dao

Purpose: Manages proposals, enforces classed quorums, vote caps, timelocks, and handles the routed execution of proposals.

- **Accounts (PDAs):**
 - **ProposalPDA(id):** Stores information about the proposal, including class, target programs, calldata, execution time (ETA), and state.
 - **BallotPDA(proposal, voter):** Stores the snapshot of counted weight (post-cap) for each voter.
- **Key Instructions:**
 - **propose(class, targets[], calldata[]):** Submits a proposal to the gateway for validation and voting.
 - **cast_vote(voter, weight):** Records a vote for a given proposal, supports both identity-only and hybrid HII-weighted modes.
 - **queue(id) / execute(id):** Enqueues proposals for execution after the required timelock; routes calls to target programs.
- **Guards:**
 - **Per-identity cap:** Enforces a 0.1% cap on voting weight per identity.
 - **Minimum participation for C3 proposals:** Ensures quorum is met for foundational changes.
 - **Veto-less design:** Supermajorities may be required for certain decisions, but no single veto power exists.
- **Events:**
 - **ProposalCreated, VoteCast, Queued, Executed, ConfigUpdated.**

B) Config Registry — config

Purpose: Serves as the canonical on-chain parameter store, tracking all system settings with versioning and timelock mechanisms to prevent unauthorized changes.

- **Accounts:**

- **ConfigRoot:** The root configuration account.
- **ConfigVersionPDA(version):** Stores versioned parameter changes, with associated metadata such as proposal ID, author, timestamp, and ETA.

- **Key Instructions:**

- **Set (key, value):** Updates a specified parameter with a new value.
- **Promote (version):** Promotes a version to the current active configuration.
- **freeze():** Locks the parameters in an emergency, preventing further changes.

C) Treasury / Buyback Controller — treasury

Purpose: Manages treasury assets with a surplus-only buyback-and-burn mechanism, applying hygiene checks before executing any transaction.

- **Accounts:**

- **TreasuryVault:** Holds the treasury assets.
- **BuybackPDA(window_id):** Manages buyback operations with relevant settings, such as TWAP window, oracle max age, coverage threshold, and notional limits.

- **Key Instructions:**

- **queue_buyback(window, notional):** Queues a buyback for execution, subject to profit and coverage checks.
- **execute_buyback(window):** Executes a buyback in a defined window and burns tokens immediately.
- **file_report(month, ipfs_hash):** Publishes a report of the buyback action with an IPFS hash for verifiability.

- **Guards:**

- **Profit-only:** Ensures buybacks occur only when there is realized profit.
- **Coverage months:** Verifies that the system has sufficient runway before allowing buybacks.
- **Oracle freshness:** Ensures the Oracle data is up to date before any buyback actions.

D) Peace Store — store

Purpose: Manages merchant listings, orders, HII-tiered discounts, anti-wash controls, and fee distribution to the treasury, hosts, and eco funds.

- **Treasury Tie-In:** All Peace Store fees are routed to the treasury, with only realized profits eligible for buybacks, maintaining system discipline.

4.3 Governance Classes, Quorums, Timelocks Encoded

Proposals are categorized by their impact and risk, and each class enforces a different quorum and timelock requirement:

Class	Scope	Examples	Quorum	Timelock
C1 Operational	Routine, non-privileged operations	Publish manifests, rotate non-privileged keys	10%	12h
C2 Parameterized	Adjustments to bounded knobs	TWAP, freshness cap, unlock caps	30%	48h
C3 Constitutional	Foundational changes (supply, identity)	Identity cap, supply rules, upgrades, LP policy	60%	96h

Gateway pre-vote ensures any proposal exceeding the bounds of these classes is rejected before reaching the vote.

Terminology — Proposal Classes, Quorums, Timelocks

Term	Meaning	Notes
C1 Operational	Low-risk, non-privileged changes.	Requires the lowest quorum (10%) and timelock (12h).
C2 Parameterized	Bound adjustments to system parameters.	Medium quorum (30%) and timelock (48h).
C3 Constitutional	Fundamental changes to system rules.	High quorum (60%) and timelock (96h).

4.4 Voting Math & Identity Cap

Eligibility is determined by the Peace Passport, where each verified identity has a **voting cap** set at **0.1%** of the total supply. Votes are aggregated, but the total weight assigned to each identity cannot exceed the cap. Delegation is allowed, but the cap still applies post-aggregation, and delegations auto-expire every quarter unless renewed.

Votes are counted based on the **class quorum** and the majority of **counted weight** (post-delegation). For **C3 Constitutional proposals**, a **supermajority** may be required for approval.

Terminology — Voting & Identity Cap

Term	Meaning	Notes
Vote cap	Maximum allowed voting weight per identity.	Set at 0.1% (10 basis points).
Delegation	The ability to transfer voting power to another participant.	Bound by the same per-identity cap.
Supermajority	A majority greater than 50%, required for specific C3 actions.	Enhances safety for high-impact proposals.

4.5 Monetary Controls Wired Into Contracts

PS's **monetary controls** focus on **mechanized scarcity** to preserve token value over time. The total supply is capped at **10B** tokens, and the mint authority is locked at genesis. **Claim-time burn** is set at **0.75%**, ensuring that each reward claim reduces the circulating supply.

Surplus-only buybacks are governed by strict parameters, including TWAP, quote freshness, and deviation guards. The buyback process is **fail-closed**—if any check fails (e.g., stale data or excessive slippage), the system takes **no action**. The entire cycle—from missions to surplus, buyback, and burn—remains tightly coupled to prevent excessive inflation or deflation.

Terminology — Monetary Controls

Term	Meaning	Notes
Claim-time burn	Reduction in supply when rewards are claimed.	0.75% per reward claim.
Surplus-only buybacks	Buybacks executed only from realized profit.	Governed by TWAP and freshness checks.

4.6 Events & "Receipts Everywhere"

All material actions in the PS system emit **standardized, machine-verifiable receipts** that ensure full traceability. Key events, such as proposal creation, vote casting, buybacks, and identity actions, are logged and accessible. A **monthly signed digest** is generated, summarizing key operations (e.g., circulating supply, buyback logs, governance actions). Each receipt contains fields such as proposal ID, actor, action timestamp, and status, allowing complete post-execution verification.

4.7 Oracle Integration & TWAP Hygiene (Pyth + Switchboard)

PS uses **Pyth** for primary oracle data and **Switchboard** for redundancy. Oracles feed into a **TWAP window** for price smoothing and are guarded by **freshness checks**, which ensure data is up-to-date within a maximum age of **120 seconds**. If any hygiene guard fails, the controller **fails-closed** and no action is taken.

Terminology — Oracle Integration

Term	Meaning	Notes
TWAP	Time-weighted average price.	Smoothing mechanism to reduce price manipulation.
Freshness checks	Validation that the price data is current.	≤ 120 seconds to ensure accuracy.
Fail-closed	No action is taken if the guardrails fail.	Protects the treasury from inaccurate data.

4.8 Upgrades & Custody

PS uses **BPF-Upgradeable Loader** for program upgrades, with the upgrade authority held by a **DAO-controlled key/multisig**. The upgrade authority can only be rotated through **C3 Constitutional proposals** to maintain transparency and prevent undue influence. Future upgrades must be staged and well-tested before deployment.

For **custody**, sensitive contracts like governance and treasury management have defined **rotation policies** that are governed by **SPL governance** or multisig protocols. Rotation policies ensure that no single party can control sensitive keys for an extended period.

4.9 Annex A — Parameter Registry

A detailed parameter registry defines every key in the system, its type, range, default value, and the class of proposal required for changes. Critical keys such as the **vote cap**, **buyback thresholds**, and **supply burn rate** are set to **C3 Constitutional** to ensure they can only be altered with significant governance involvement.

Terminology — Parameter Registry

Key	Type	Range / Policy	Default	Class
gov.vote_cap_per_id_bps	u16	5–20 (0.05%–0.20%)	10	C3

buyback.coverage_threshold_months	u8	3–24	12	C3
gov.timelock.operational_s	u32	≥ 43,200	43,200	C3
rewards.claim_burn_bps	u16	Fixed	75	C3
identity.credential_ttl_days	u16	90–365	180	C2

Cross-references: For full details on the parameter registry and its operational relationships, refer to **Appendix B** (Parameter Annex) and **Appendix K** (Code & Schemas).

4.10 Peace Passport & Privacy

The **Peace Passport** serves as a critical identity verification tool in the PS ecosystem, allowing one **cNFT per verified human**. This **non-transferable** token provides a unique, auditable identity layer while minimizing exposure to Personally Identifiable Information (PII). Rather than storing complete personal data, the Peace Passport embeds **proofs** and **flags** in its metadata, providing only the necessary information to enable participation in the DAO.

Governance controls are integrated into the passport revocation process, with revocations managed through governance Standard Operating Procedures (SOPs). Retention of identity data is carefully controlled through defined **default periods** (such as **90-180 days**) and, when expired, is automatically purged to minimize long-term data storage. The PS system enforces privacy through the **minimal retention** principle, ensuring that identity information is only stored as long as necessary for governance participation, and no personal data is retained unnecessarily.

Terminology — Peace Passport & Privacy

Term	Meaning	Notes
cNFT (Certified NFT)	Non-transferable NFT linked to a verified identity.	Minimal PII stored in metadata.
Revocation SOP	Governance process for identity invalidation.	Enforces integrity and privacy.

Minimal PII	Storing only the minimum necessary information for identity verification.	Privacy-by-design approach.
Purge path	Automated removal of expired identity data.	Ensures compliance with retention policies.

4.11 Commerce Plane (Store) → Treasury → Buyback Loop

The **Peace Store** functions as the commerce hub, with **discount logic tied to Human Impact Index (HII) tiers**. This enables a dynamic reward system where customers receive discounts based on their verified contributions to global peace efforts. To prevent system abuse, anti-wash filters are integrated to ensure that transaction data remains clean and free from manipulation. Disputes within the system are handled according to a defined service-level agreement (SLA), ensuring a timely resolution.

All transaction fees are automatically split between the **Treasury, host accounts, and eco funds**, contributing to the sustainability of the ecosystem. **Clean GMV (Gross Merchandise Value)** and **Net Contribution** are defined to ensure accounting clarity, ensuring that only **realized profits** contribute to buybacks. This creates a robust and auditable system that ties commerce directly to treasury operations and ultimately to buybacks, reinforcing the system’s mechanized scarcity design.

Terminology — Commerce Plane

Term	Meaning	Notes
HII tiers	Human Impact Index-based discount levels.	Tied to verified contributions.
Anti-wash filters	Mechanism to detect and prevent fraudulent transaction activity.	Ensures fair and clean accounting.
Clean GMV	Total transaction value excluding invalid or suspicious orders.	Used to calculate accurate revenue.

Net Contribution	The portion of GMV that contributes to the ecosystem after deductions.	Realized profit routed to buybacks.
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4.12 Off-Chain Perimeter & Data Plane

The **off-chain perimeter** is managed by an **Indexer** and **KPI service**, which aggregates and organizes data into canonical tables. These tables track key metrics, including **missions, rewards, buybacks, de-washed GMV**, and **governance concentration**. Monthly signed reports are **anchored on-chain** for verification and transparency, ensuring that all data used for decision-making is auditable and transparent.

In addition, **forums** and **PSIP templates** serve as the primary channels for off-chain discussions, proposal submissions, and formal voting. The **delegate registry** tracks statements of purpose, voting history, and potential conflicts of interest for transparency. Proposals submitted via the gateway include **disclosure and conflict of interest (COI) attestations** as part of the proposal checks.

Terminology — Off-Chain Perimeter & Data Plane

Term	Meaning	Notes
Indexer	Off-chain service that aggregates key metrics.	Provides a live overview of system health.
KPI service	System that tracks and organizes key performance indicators.	Anchors reports on-chain for integrity.
PSIP	Peace Soldiers Improvement Proposal template.	Standardized format for proposal submissions.
Delegate registry	Record of delegate statements, voting history, and COI.	Enhances transparency and accountability.

4.13 Security Posture Defense

The PS DAO implements a multi-layered security posture to defend against common vulnerabilities and risks in decentralized governance. Key measures include:

- **Market hygiene:** Ensuring market data integrity with **TWAP (Time-Weighted Average Price)**, **maximum age** of quotes, **deviation checks**, **notional caps**, and **coverage floors**. Any failure of these hygiene checks triggers a **fail-closed** action, where no further actions (such as buybacks) are executed until the data is verified.
- **Governance capture protection:** The DAO combats governance capture via a **per-identity voting cap** (0.1%) and optional **HII-weighted** voting. **Classed quorums** and **timelocks** further ensure that changes to the DAO require broad community support and are not easily manipulated.
- **Data exposure control:** To minimize the risk of exposing PII (Personally Identifiable Information), PS uses **minimal PII retention**, alongside **encryption**, **role-based access control (RBAC)**, and **deletion paths** to manage data securely. **Public digests** and **KPI proofs** ensure transparency and can be verified by external parties.

Terminology — Security Posture

Term	Meaning	Notes
TWAP	Time-weighted average price for market stability.	Protects against price manipulation.
Fail-closed	No action taken if a check fails.	Ensures safety in case of data issues.
Per-identity cap	Limits voting power to 0.1% per identity.	Prevents governance centralization.
Minimal PII	Collects only necessary identity data.	Enhances privacy and minimizes risk.
RBAC	Role-based access control for data security.	Limits access based on predefined roles.

4.14 Interoperability with DAO-IDEAS Modules

PS plans to pilot **DAO-IDEAS modules** that enhance governance flexibility while maintaining safety. These include:

- **AI-assisted reputation shards (SBTs)**: Reputation is aggregated from governance participation and activity, with **non-transferable** scores that decay over time. These are used to **boost attention** and **triage delegation leaderboards**.
- **Fractal Sub-DAOs**: These smaller DAOs inherit PS's core rules but operate within defined bounds, with resource flows controlled by **membranes**. These sub-DAOs undergo **quarterly health checks** and are supported by **C3 charters** and **C2 budgets**.
- **Liquid-merit delegation**: Stewards stake PS tokens for skin-in-the-game, and their performance is tracked on **public leaderboards**. **Quadratic dampening** reduces the influence of large delegations to ensure fairness.

These modules are piloted **behind C2 (parameterized) proposals**, ensuring that they remain **subordinate** to PS's foundational governance principles.

Terminology — DAO-IDEAS Modules

Term	Meaning	Notes
Reputation shards (SBTs)	Non-transferable reputation scores.	Decay over time, boosting governance influence.
Fractal sub-DAOs	Smaller DAOs within PS, inheriting core rules.	Governed within the bounds of PS's Config Registry.
Liquid-merit delegation	Delegation based on performance and stake.	Public KPIs and quadratic dampening to prevent capture.

4.15 Deployment & Operations Blueprint (Solana)

Deployment Roadmap:

1. **Devnet → Shadow-fork testing**: Program IDs, PDAs, fixture setups, and oracle simulators will be deployed for testing.

2. **Pre-mainnet checklist:** Includes **unit/property tests**, **fuzzing**, **economic simulations**, and formal parameter entry into **config@v1**. **Council keys** will be generated, and **upgrade authority** custody will be established.
3. **Mainnet-beta:** Set critical defaults (e.g., **burn_bps=75**, **coverage floor**, **buybacks OFF** for 60 days). The **custody registry** and key addresses will be published.
4. **Observability:** The public dashboard will track **circulating supply**, **coverage months**, **TVDR**, **Clean GMV**, **buyback logs**, and **governance concentration**.
5. **Progressive decentralization:** In this phase, the **proposer set** will be expanded, **steward elections** will be held, and modules like **reputation** and **sub-DAOs** will be piloted behind timelocks.

Terminology — Deployment & Ops

Term	Meaning	Notes
Devnet	Initial deployment network for testing.	Mimics mainnet with minimal risk.
Shadow-fork	Private copy of mainnet for testing changes.	Validates real-world impact without risk.
Mainnet-beta	Limited mainnet launch for early real-world testing.	Critical parameters set, custody registry published.
Progressive decentralization	Gradual transition of authority to the community.	Expands proposer set and holds steward elections.

This architecture provides a robust, modular, and secure framework for the PS DAO, from initial testing and deployment to ongoing governance and system evolution. Each layer is carefully constructed with **security** and **efficiency** in mind, ensuring PS remains a resilient, scalable ecosystem that aligns with its mission of promoting verified human impact through decentralized decision-making.

4.16 External Precedent for Phased Governance

There are moments when the most responsible choice is to **sequence** community voting rather than open every surface at once. The correct approach is not to use ad-hoc pauses, but to pre-commit milestones, KPIs, and timelines in the Config Registry, allowing everyone to see the conditions for opening or widening governance. PS adopts this posture explicitly: if voting on certain classes is phased over a period, the Registry will clearly outline the objective gates (for example, minimum participation targets, coverage-month floors, and audit sign-offs) and the corresponding calendar **windows** for review. This makes a temporary constraint both **transparent and reversible**, thereby avoiding the uncertainty that follows from informal announcements. The lesson is reinforced by other ecosystems that staged governance while shipping core infrastructure: the projects that fared best wrote their criteria down, hashed them on-chain, and stuck to them. *(Chronology and public statements are summarized in the body and expanded in **Annex C-7** for auditors.)*

Terminology — Phased Governance

Term	Meaning	Notes
Pre-committed milestone	A published, parameterized condition to advance decentralization.	Stored in the Config Registry with a hash and ETA.
KPI gate	An objective metric (e.g., quorum hit-rate, coverage floor) required before opening a surface.	Evaluated in signed monthly digests.
Time-boxed phase	A defined window after which status is re-evaluated on-chain.	Prevents indefinite “temporary” pauses.

Part 4 — Deliverables Recap

Part 4 delivers a **concrete, testable blueprint** for PS on Solana. It defines the program suite (governance gateway, Config Registry, treasury controller, Peace Store, vesting, indexers) and the **IDL-level surfaces** those programs expose. It wires **bounded parameters** to **governance classes** so that quorums and timelocks protect the right levers, and it specifies **oracle/TWAP hygiene** for fail-closed buybacks. It documents the **Peace Store → Treasury → Buyback** loop with **audit-grade receipts and KPIs**, ensuring that commerce flows are observable end-to-end. Finally, it outlines the upgrade authority and custody model, consistent with modern Realms/ SPL Governance practice, including rotation rules and the publication of an address & custody registry.

In short: you now have the Solana program/IDL map, accounts/PDAs and instructions, the parameter registry excerpt and bindings, the oracle hygiene rails and fail-closed controller behavior, the Store→Treasury loop with receipts, and a custody model that keeps upgrade power in the DAO—nothing ad-hoc, nothing opaque.

Terminology — Part 4 Recap

Term	Meaning	Notes
IDL	Interface definition that fixes program instructions/accounts.	Basis for client correctness and audits.
Fail-closed	Controller does nothing if any guard fails.	Preferred behavior for treasury actions.
Custody registry	Canon of program IDs, upgrade authorities, vaults, and rotation policy.	Governance-gated modifications only.

Part 5 — Execution Roadmap, Risk & Legal, and Operational Appendices

5.1 Phased Execution Roadmap

Phase 0 — Governance & Treasury “Safety-On” Bootstrap.

The first phase ships a **minimal, safe, observable baseline**: classed voting, Config Registry v1, treasury hygiene in **monitor-only** mode, and public KPIs. We promote **Config v1** with explicit defaults, publish the diff and ETA, and **freeze Critical keys** (upgrades, buyback toggles, oracle max-age, coverage threshold) to prevent accidental drift. Rewards are parameterized at **claim_burn_bps = 75 (0.75%)**, and **coverage months are displayed** on the public dashboard. Buybacks remain **OFF** while we observe live telemetry for **60 days**, building confidence in data quality and gateway discipline. Proposal hygiene goes live with **single-subject checklists, disclosure templates, and rollback boilerplates**. The first Signed Monthly Digest is published, accompanied by an on-chain hash. Governance, incentives/treasury, and data/observability pods share the workload. The phase ends when Config v1 is active,

Digest #1 is live, classed voting is functioning, and the treasury controller has run successfully in monitor mode.

Phase 1 — Identity, Missions & Store MVP.

Next, we connect **identity to action**. The program supports **accrual** → **claim** with the **0.75% claim-burn**, and we launch three **baseline mission templates** (education, volunteering, activity) with **audit sampling** and **appeal windows**. Identity issuance stabilizes under low fraud flags, while the host's **reputation** begins to accumulate. The Store ships as an MVP with **merchant onboarding**, **H11-tiered discounts**, **anti-wash guards**, and a **fee splitter** to treasury and partners. **Epoch ceilings** prevent reward overruns, and a policy **discount cap** $\leq 30\%$ keeps incentives productive. The phase closes when we record the first **Clean-GMV readings**, publish **Digest #2**, and confirm stable issuance of identities.

Phase 2 — Treasury Buyback Controller.

With telemetry in place, we enable surplus-only buybacks behind the complete set of hygiene rails: Oracle **max-age**, **TWAP** window, **per-window notional cap**, **deviation fence**, and a **coverage-months floor**. Any failed check results in a **NOOP**; there are no exception paths. The Store routes realized a surplus for the treasury, and successful buybacks are burned **immediately**. Exit requires **two non-consecutive** successful buyback windows with complete receipts and **zero hygiene violations**, proving the controller behaves predictably under varying conditions.

Phase 3 — Progressive Decentralization & Pilots.

Authority expands as evidence accumulates. The **proposer set** widens, **steward elections** begin, and we pilot DAO-IDEAS features **behind timelocks**. **Liquid-merit delegation** goes first with **auto-expiry**, optional **steward self-stake**, and **quadratic dampening** to avoid popularity capture. **AI-assisted reputation shards (SBTs)** run on the testnet with multi-oracle inputs and an appeal window before any mainnet flag is set. **Fractal sub-DAOs** receive C3 charters and C2 budgets, with **quarterly sync votes** and **auto-dissolution** for inactivity. The phase ends when a steward board is seated, one or two pilots have completed their **post-mortems**, and **concentration metrics** remain stable.

Phase 4 — Maturity & Hardening.

The final phase focuses on durability. After **external audits**, we may **finalize** critical programs by **revoking upgrade authority**, retaining the C3 path for constitutional upgrades only. We publish a **Document Pack**, which includes Terms of Use, Privacy, Merchant Agreement, Open Methodology, and Risk Factors, with content hashes anchored on-chain. The Store broadens categories under **dispute SLAs** and a **chargeback reserves** policy. Exit requires **audit sign-offs**, the Document Pack v1 public, and a **run-rate Clean GMV** sufficient to support policy buybacks without stressing coverage floors.

Terminology — Execution Roadmap

Term	Meaning	Notes
Monitor mode	Controller computes decisions but does not execute.	Used in Phase 0 to validate hygiene.
Epoch ceiling	Upper bound on rewards within a period.	Prevents budget overruns.
Non-consecutive windows	Separated successful executions proving stability over time.	Reduces coincidence risk.
Finalize	Revoke upgrade authority after audit.	Keeps constitutional upgrade path only.

Cross-references: Phase gates and defaults are summarized here and specified in **Appendix B (Parameter Annex)**, **Appendix D (Operational Checklists)**, and **Appendix H (Governance Communications & Monthly Digest Schema)**.

5.2 Dependencies & Resourcing

We organize execution around **pods that own surfaces end-to-end**—Access, Identity, Proof, Incentives, Commerce, Governance, and Data—so there is a single locus of accountability from design to on-chain change and post-deployment observability. Each pod has a named **DRI** who owns the scope, schedule, and quality. Cross-cutting Risk Councils—Security/Privacy, Treasury, and Compliance—set guardrails, review diffs before they queue, and publish short written sign-offs that are attached to proposals. The operating cadence is deliberately tight: **weekly sprints**, **bi-weekly demos** across pods, and a **monthly Signed Digest** that becomes a gating artifact for continued funding, pilot promotion, and parameter elevation. This rhythm keeps the feedback loop short while anchoring material decisions in artifacts that auditors can recompute.

Terminology — Resourcing & Cadence

Term	Meaning	Notes
Pod	Cross-functional team that owns a product surface end-to-end.	Design → code → dashboards.
DRI	Directly Responsible Individual for scope and outcomes.	Named in each proposal.
Risk Council	Security/Privacy, Treasury, or Compliance reviewers.	Guardrails + written sign-offs.
Signed Digest	Monthly KPI bundle with on-chain hash.	Gating artifact for funding/pilots.

5.3 Risk Register

PS treats risk as a set of **testable hypotheses** with controls and public evidence. Market and oracle risk is contained by **TWAP alignment**, **quote-freshness limits**, **per-window notional caps**, and a **circuit breaker** that halts the controller when hygiene fails; receipts prove that halts were automatic rather than discretionary. Treasury risk is curbed by a **profit-only rule** and a **coverage-months floor**, with buyback activity visible in monthly digests. Privacy risk is minimized by **keeping PII off-chain**, applying **RBAC**, and operating deletion paths under the discipline of DPIA/ROPA. Governance capture is addressed with a **0.1% (10 bps) per-identity cap**, **classed quorums and timelocks**, concentration KPIs, and **delegation auto-expiry**. Emissions risk during demand droughts is managed with **epoch ceilings**, coverage-first levers, and conservative multipliers. Multi-signal verification, random audits, and anti-wash filters with clear evidence SLAs limit fraud and wash activity. Liquidity shocks are mitigated through an adaptive liquidity playbook, mirrored pools, and LP locks of 12 months or longer. Operational security focuses on **C3-gated upgrades**, external audits, published custody registry, and **MPC/HSM** operations. Jurisdictional variance is handled through versioned legal documents and, where applicable, a **DAO LLC hybrid** for off-chain interfaces—without compromising on-chain supremacy.

Risk Register — At a Glance

Category	Risk	Control & Evidence
Market/Oracle	Stale/thin prices distort buybacks	TWAP, freshness, notional caps, circuit breaker; fail-closed controller; receipts & digest entries.
Treasury	Optics-driven buybacks deplete reserves	Profit-only rule, coverage floor; public receipts & monthly digest.
Data/Privacy	PII exposure	PII off-chain + RBAC; DPIA/ROPA; deletion paths.
Governance	Capture by whales/delegates	0.1% cap; classed quorums/timelocks; concentration KPIs; auto-expiry delegation.
Emissions	Reflexive rewards in droughts	Epoch ceilings; coverage-first levers; bounded multipliers.
Fraud/Wash	Mission farming / Store wash	Multi-signal verification; random audits; anti-wash filters; evidence SLAs.
Liquidity	LP depth shocks/volatility	≥12m LP locks; mirrored pools; liquidity playbook.
Ops/Security	Upgrade/custody failures	C3 upgrades; external audits; custody registry; MPC/HSM.
Legal/Compliance	Jurisdictional variance	Versioned Terms/Privacy/Merchant; DAO LLC hybrid if needed.

5.4 Legal & Policy Workstream

The legal wrapper, where used, is **non-discretionary and ministerial**: it signs on behalf of the DAO when a signature is unavoidable (e.g., vendor onboarding or narrow KYC escrow) and maintains records. It does **not** direct treasury flows or override on-chain outcomes. We publish a **Document Pack v1.0**, which includes the Terms of Use, Privacy Notice, Merchant Agreement, Open Methodology, and Risk Factors, under semantic versioning with content hashes anchored on-chain, allowing changes to be visible and attributable. In jurisdictions where it reduces friction, PS may adopt a **DAO-LLC hybrid for off-chain interfaces; even then, enforcement remains on-chain, protected** by timelocks and receipts. This split preserves clarity for partners and regulators without compromising decentralization.

Terminology — Legal & Policy

Term	Meaning	Notes
On-chain supremacy	Code-governed actions outrank off-chain bodies.	Wrapper cannot move funds.
Document Pack	Canonical policy documents with on-chain hashes.	Publicly diff-able.
DAO LLC hybrid	Limited legal shell for interfaces only.	No treasury discretion.

5.5 KPI Dictionary (Dashboard & Digest)

KPIs are defined for **recomputation by third parties**. **Coverage Months** measures runway at current accrual and parameters and gates buybacks/APR decisions. **Clean GMV** is GMV after de-wash (invalid/suspicious orders removed) and drives surplus estimates. **TVDR** (Token Velocity / Demand Ratio) and **Governance Concentration** (Herfindahl/Gini) characterize monetary use and influence dispersion. **Buyback Hygiene** reports the pass-rate of TWAP/freshness/deviation checks, with program receipts cross-checked against treasury reports. **Mission Integrity** captures audit-sampling rates, appeal SLAs, and anomaly heatmaps. Each figure appears in the dashboard and the **Signed Monthly Digest**, with links to the underlying tables.

KPI Summary

KPI	Definition	Evidence
Coverage Months	Runway at smoothed net outflows under current params.	Dashboard tile; digest value; controller receipts.
Clean GMV	GMV net of flags/invalidations post de-wash.	Store pipeline logs; digest reconciliation.
TVDR	Velocity/demand indicator for token utility.	Methodology note; series in dashboard.
Concentration	Herfindahl & Gini for voting power.	Snapshot calc; monthly series.
Buyback Hygiene	% windows passing TWAP/freshness/deviation.	Controller receipts; oracle logs.
Mission Integrity	Sampling rate, appeal SLA, anomaly map.	Proof tables; adjudication receipts.

5.6 Receipt Schemas (Standardized, Machine-Verifiable)

Every privileged action emits a **receipt with canonical fields**, enabling independent reconstruction of events. Schemas are stable and versioned; JSON/IDLs live in **Appendix E**.

- **gov_execute** — fields: `proposal_id`, `class`, `quorum_bps`, `eta`, `targets[]`, `calldata_hash`, `signer_set`, `txid`, `config_version`.
- **buyback_execute** — fields: `window_id`, `twap_window_s`, `oracle_age_s`, `notional`, `base_asset`, `burn_txids[]`, `proof_hash`.
- **reward_claimed** — fields: `user`, `epoch`, `gross_units`, `burn_bps`, `burn_amount`, `net_amount`, `txid`.

- **store_order** — fields: `order_id`, `merchant_id`, `amount`, `discount_bps`, `flags[]`, `settlement_eta`.

All receipts are summarized monthly in a **Signed Digest** with an **on-chain content hash** and links to per-action logs.

Terminology — Receipts

Term	Meaning	Notes
Calldata hash	Hash of executed payload.	Binds action to approved content.
Proof hash	Reference to supporting data (e.g., buyback proof).	Stored off-chain, anchored on-chain.

5.7 Custody & Address Registry Template

We publish a **Custody & Address Registry** post-audit, listing all **program IDs** (dao, config, treasury, rewards, missions, store) with deployment transactions; **authorities** (upgrade authority held by DAO multisig/realm, council signers, reporter keys) and their **rotation policy** (C3-gated); and **vaults** (treasury vaults, LP lock accounts with **≥12-month** locks, buyback PDAs). Updates to this registry are **governance-gated**, and every rotation event emits a receipt, creating an immutable custody trail for auditors.

Registry Outline

Section	Contents	Governance Notes
Programs	Program IDs + deploy txids	Changes via C3 only.
Authorities	Upgrade/council/reporter keys	Rotation policy encoded; receipts emitted.
Vaults	Treasury, LP locks, buyback PDAs	Lock terms visible; unlocks receipted.

5.8 SOPs

EmergencyPause (governance-scoped, ≤24h). When objective triggers fire—such as oracle staleness or deviation beyond policy—the system can activate a narrowly scoped **Emergency Pause for up to 24** hours. Within **two hours**, a status page and an in-app banner are published. The full trail is received and summarized in the following digest. The pause cannot seize funds already time-locked; it only halts new scheduling.

Proposal Hygiene. Every submission must conform to the **single-subject rule**, include a **disclosure pack** and **rollback plan**, and pass **duplicate suppression** at the gateway. Hygiene raises proposal quality, reduces voter fatigue, and shortens auditor review time.

Release Gates. No mainnet execution proceeds without **unit/property tests**, **fuzzing**, **economic simulations**, and **canary runs**. Each month, the team links proofs and receipts in the digest, allowing reviewers to spot-check real artifacts rather than relying solely on assurances.

Terminology — SOPs

Term	Meaning	Notes
EmergencyPause	Short, scope-limited halt under clear triggers.	Auto-expires; receipt-bound.
Single-subject	One coherent change per proposal.	Enforced at gateway.
Canary run	Low-blast-radius production test.	Required before broad enablement.

5.9 Economics Tie-Out — Why This Roadmap Preserves Value

The roadmap converts tokenomics into a **mechanized discipline**. A hard **10B** ceiling and a **0.75% claim-time burn** embed scarcity at the moment verified rewards enter circulation. **Surplus-only buybacks**, executed only when hygiene checks pass, prevent discretionary pumps and align value accrual with real surplus. The loop is closed and observable: **Missions** → **Claim (−0.75%)** → **Store** → **Surplus** → **Guarded Buybacks** → **Burn**. Deflationary pressure emerges only when **Clean GMV** scales sustainably, allowing price support to track productive activity rather than speculation. By keeping treasury actions **fail-closed** and binding every change to receipts and public KPIs, the system preserves solvency first and builds credibility over time.

Terminology — Economics

Term	Meaning	Notes
Mechanized scarcity	Rule-driven burn + surplus-only buybacks.	No discretionary emissions.
Clean GMV	GMV after de-wash pipeline.	Drives surplus estimates.
Fail-closed	Do nothing when hygiene fails.	Preserves reserves.

Part 5 — Deliverables Recap

This part provides a **date-driven execution plan** (Day 0 → Day 365) with crisp exit criteria, resourcing, and dependencies; a **risk register** paired with on-chain controls, SLAs, and public evidence; a **legal/documentation plan** that preserves on-chain supremacy while minimizing off-chain discretion; and operational appendices: a **KPI dictionary**, **receipt schemas**, a **custody & address registry template**, and **SOPs** that make the path from proposal to production **auditable by design**.

Part 6 — Appendices

Appendix A — KPI Dictionary (definitions, formulas, cadence, ownership)

These KPIs are the public, recomputable gauges of PS DAO health. Each metric has a plain-English definition, a precise calculation method, a publishing cadence, and an accountable owner. All results are displayed on the public dashboard and compiled monthly into a Signed Digest, whose content hash is anchored on-chain. Where a formula could be sensitive to short-term noise, we use smoothed (rolling/EWMA) inputs and exclude locked or illiquid balances.

KPI	Definition (source)	Formula / Method	Cadence	Owner
Coverage Months	Reward runway at current accrual & parameters; gate for buybacks/APR	<code>coverage_months = treasury_reserve / monthly_net_outflows</code> , where <i>monthly_net_outflows</i> is a 30-day EWMA of net programmatic outflows excluding locked/illiquid funds. Buybacks are blocked if <code>coverage_months < policy_floor</code> .	Monthly (Signed Digest)	Treasury Pod
Clean GMV	GMV net of flagged/invalid orders after the de-wash pipeline; drives surplus estimates.	<code>clean_gmv = Σ settled_orders - Σ invalidated_or_flagged_orders</code> (risk-weighted exclusions).	Weekly dashboard; monthly digest	Store Pod
Net Contribution	Portion of commerce that actually accrues to treasury after	<code>net_contribution = clean_gmv $\times$ (BaseFee -</code>	Monthly	Store + Treasury

	policy reserves.	$\text{DiscountSubsidy}) - \text{Risk\&OpsReserves.}$		
Buyback Hygiene Pass-Rate	Share of windows in which all hygiene checks pass.	$\text{pass_rate} = \frac{\text{passed_windows}}{\text{total_windows}}$, where a pass requires TWAP , freshness , and deviation checks to pass; controller is fail-closed on any failure.	Monthly	Treasury
Circulating Supply Δ	Net annualized supply change by scenario.	$\Delta\text{circ} = \text{Mint} - \text{ClaimBurn} - \text{BuybackBurn}$; ClaimBurn is 0.75% of eligible reward at claim.	Monthly	Data
Governance Concentration	Dispersion of counted vote weight; capture monitor.	Standard HHI and Gini computed on post-cap (0.1%) weights per snapshot.	Monthly	Governance
Identity Integrity	Health of the identity set and adjudication pipeline.	% challenged , % revoked , appeal SLA compliance (median/95th).	Monthly	Access
Digest Integrity	Verifiability of public reporting.	Verify digest content hash against published report JSON; sample receipts.	Monthly	Data

Methodological notes (auditor-friendly):

- *Coverage Months* uses a smoothed outflow estimate to avoid one-off spikes biasing gating decisions.
- *Clean GMV* removes refunds, test orders, and risk-flagged items via deterministic rules; exclusions are listed in the digest.
- *Buyback Hygiene* counts a window as a pass only if **all** rails pass (TWAP alignment,

quote freshness $\leq$ policy, deviation $\leq$ fence); any single failure $\rightarrow$ **NOOP**.

- *Governance Concentration* is computed on **counted** weights (after the 0.1% per-identity cap and any quadratic dampening, if enabled).

Appendix B — Parameter Annex

The **Config Registry** is the single on-chain source of truth. Each promotion writes an **append-only diff** with proposal ID and ETA, so parameter history is transparent and revertible. Classes indicate the governance path: **C2 (Parameterized)** for bounded tuning and **C3 (Constitutional)** for systemic levers. Defaults load at `config@v1` and can only move **within** stated ranges and classes.

Key	Type	Range / Policy	Default	Class
<code>supply.max</code>	u64	fixed	10,000,000,000	C3
<code>rewards.claim_burn_bps</code>	u16	fixed at 75 bps	75	C3
<code>buyback.funding_source</code>	enum	<code>surplus_only</code>	<code>surplus_only</code>	C3
<code>buyback.twap_window_s</code>	u32	600–21,600	7,200	C2
<code>buyback.oracle_max_age_s</code>	u32	30–600	120	C2
<code>gov.vote_cap_per_id_bps</code>	u16	5–20 ($\rightarrow$ 0.05%–0.20%)	10 (0.1%)	C3
<code>gov.quorum.operational_bps</code>	u16	500–2,000	1,000 (10%)	C3

gov.quorum.economic_bps	u16	2,000–4,000	3,000 (30%)	C3
gov.quorum.critical_bps	u16	5,000–7,000	6,000 (60%)	C3
gov.timelock.operational_s	u32	≥ 43,200 (12h)	43,200	C3
gov.timelock.economic_s	u32	≥ 172,800 (48h)	172,800	C3
gov.timelock.critical_s	u32	≥ 345,600 (96h)	345,600	C3
liquidity.lp_lock_months	u16	≥ 12	12	C3
store.discount_cap_bps	u16	≤ 3,000 (≤30%)	≤ 3,000	C2
identity.credential_ttl_days	u16	90–365	180	C2 (policy)
buyback.coverage_threshold_months	u8	3–24	12	C3 (policy, gating)

Standards & conventions:

- **Basis points (bps):** 10 bps = **0.1%**. The per-identity vote cap is expressed as `gov.vote_cap_per_id_bps = 10`.
- **Minima/Maxima:** Ranges are inclusive; timelocks are **minimum** delays.
- **Critical changes:** Any modification to **C3** keys requires the C3 path plus a rollback plan and KPI note in the proposal.
- **Gating semantics:** `buyback.coverage_threshold_months` halts buybacks automatically when runway falls below the floor; resumption requires both telemetry recovery and normal scheduling.

Cross-references: Calculation windows and KPI formulas—Appendix A; receipt schemas—Part 5 §5.6; governance classes and timelocks—Part 4 §4.3; identity/privacy defaults—Part 4 §4.10.

Appendix C — PSIP Templates

These templates keep proposals crisp, comparable, and auditable. Every PSIP is single-subject, binds execution to a content hash, and carries disclosures and a rollback plan proportional to risk class. Copy the blocks below verbatim when drafting.

C1 — PSIP (Operational)

Title / Summary

A one-sentence description of the operational change and the intended outcome.

Class

C1 — Operational (Timelock $\geq$ 12h; Quorum 10%).

Motivation / Scope (single-subject)

Explain why this change is needed now. Describe exactly one scope of work; defer extras to follow-on PSIPs.

Payload

- `targets[ ]`: program IDs or endpoints to be called
- `calldata[ ]`: ABI/IDL-encoded parameters (hash bound)
- **Expected events**: list the receipts you expect to see (e.g., `ConfigUpdated`, `gov_execute`)

Disclosures

Dependencies (upstream params, keys, or services). Include a brief rollback note describing the prior good state.

Checklist

- Schema valid
- Payload content-hash computed and included
- Bounds not applicable (C1)

Outcomes & Receipts

Reference the `gov_execute` receipt IDs post-execution and link the dashboard tile(s) updated.

C2 — PSIP (Parameterized)

Class

C2 — Parameterized (Timelock $\geq 48\text{h}$; Quorum 30%).

Parameter Changes (Annex-B keys)

Tabulate **old** → **new** values. Confirm each is **within bounds** set in the Registry.

Risk Surface

Identify whether the change touches **treasury**, **oracle/market data**, or **governance safety**.

Tests

Links to simulations, dry-run proofs (shadow-fork / devnet logs), and any backtests.

Rollback Plan

Name the pre-approved fallback **config version** and the trigger that would roll back.

Gate Checks

Affirm Registry bounds pass. The gateway **rejects** any out-of-range payload pre-vote.

C3 — PSIP (Constitutional)

Class

C3 — Constitutional (Timelock $\geq 96\text{h}$; Quorum 60%).

Change Type

Choose one: supply rules, per-identity vote cap, upgrades, LP policy, compliance mode.

Evidence Pack

Attach external **audits**, **economic simulations**, and (if relevant) a **legal memo**. Summarize the safety analysis in one page up front.

Supermajority

If required, specify the threshold (e.g., 2/3 of counted weight).

Mandatory Inclusions

- **Rollback plan** (with pre-staged config version)
- **Staged execution** (split steps across proposals if risk demands)
- **Address & custody updates** (publish diffs to the registry)

Common Blocks (all PSIPs)

COI Attestation

Proposer and endorsers disclose conflicts of interest and affirm accuracy.

Impact Analysis

List KPIs touched: **Clean GMV**, **Coverage Months**, **Governance Concentration**, etc., with expected directionality.

Execution Plan

Queue ETA, signer set, and how `dao.execute` will route calls. Note timelock class.

Post-Execution Verification

List receipt IDs (e.g., `gov_execute`, `ConfigUpdated`) and the dashboard tiles or report sections to update.

Appendix D — Checklists

Checklists make releases predictable and auditable. Each item is a yes/no gate tied to artifacts.

D.1 Release Gate (mainnet)

- Unit/property tests and fuzzing **green**
- Shadow-fork **dry run** completed, logs linked
- `config@v1` (or next) **diff** published with ETA
- Council keys verified; **custody registry** updated
- **Monthly Digest #N** queued; rollback plan signed by DRIs and Risk Council

D.2 Buyback Window

- `profit_flag == true`
- `coverage_months ≥ 12` (policy floor)
- `twap_window_s == 7,200` (default)
- `oracle_age_s ≤ 120`
- `deviation_bps ≤ fence` (policy default 75 bps; range 25–200 bps)
- `notional ≤ cap` → **execute**, else **NOOP**
- `buyback_execute` receipt emitted; included in Digest

D.3 Identity & Voting

- Passport **verified**; per-ID cap **0.1%** applied **post-delegation**
- Delegations carry **auto-expiry** date
- Snapshot taken; counted weights finalized before vote close

D.4 Store Anti-Wash

- Merchant tiering applied; **velocity thresholds** enforced
- Loop-detection rules run; flags placed where triggered
- **Discount caps** respected; settlement holds on flagged orders

D.5 Monthly Digest

- Parameters, KPI values, and **receipt IDs** compiled
- Changes since prior month noted; **content hash** anchored on-chain

- Public export (JSON) available at stable location
-

Appendix E — Receipt Schemas

All receipts are standardized and versioned. Rates are expressed in **basis points (bps)**; timestamps are UNIX seconds; IDs are content-hash bound where applicable.

gov_execute

```
{ proposal_id, class, quorum_bps, eta, hash, targets[], txid, config_version }
```

Emitted by `dao.execute` upon successful, hash-bound execution.

buyback_execute

```
{ window_id, twap_window_s, oracle_age_s, deviation_bps, notional, burns[], report_hash }
```

Emitted only if all guards (TWAP, freshness, deviation, coverage) passed; `burns[ ]` lists burn txids.

reward_claimed

```
{ user, epoch, gross_units, burn_bps:75, burn_amount, net_amount, txid }
```

Claim-time burn applied mechanically at 75 bps.

store_order

```
{ order_id, user, merchant, amount, discount_bps, flags[], settlement_eta }
```

Flags convey anti-wash outcomes and settlement holds.

Appendix F — Custody & Address Registry

The registry is the canonical map of programs, authorities, and vaults. It is published post-audit and updated only via governance.

Programs

- `dao, config, treasury, rewards, missions, store`
- Program IDs and **deploy txids** recorded

Authorities

- **Upgrade authority**: DAO multisig/realm (rotation via **C3** only)
- **Council signer set**
- **Reporter keys** (indexers, digest signers)
- Rotation policy encoded; every rotation emits a receipt

Vaults

- Treasury vault addresses
 - **LP lock accounts** (lock term ≥ 12 months)
 - Buyback PDAs
-

Appendix G — DAO-IDEAS Integrations (gated pilots)

These modules expand capability without compromising core safety. All are **C2-gated pilots** behind timelocks and subject to sunset rules.

Liquid Merit Delegation

Narrative reports and public leaderboards for stewards; optional **self-stake**; **quadratic**

dampening to soften outsized delegations; **auto-expiry** of delegations to refresh consent.

Adaptive Reputation Oracles

SBT-style shards derived from multi-oracle inputs (governance delivery, mission attestations, forum reputation), with **AI-assisted auditability** and **dispute windows**. Run on **testnet** before any mainnet flag.

Fractal Sub-DAO Membranes

Factory deployer and **boundary enforcer** to control resource/membership spillovers; **quarterly sync votes**; **auto-dissolution** on inactivity; charters via **C3**, budgets via **C2**.

Appendix H — Glossary

- **Config Registry** — The on-chain parameter store with versioning and timelocks. All adjustable policy lives here; every change leaves a diff and a receipt.
 - **Claim-Time Burn (0.75%)** — An automatic burn applied when a verified reward is claimed, reducing supply in a way tied to real human impact.
 - **Surplus-Only Buyback** — The treasury buys and burns tokens **only** from realized surplus (not optics), and only when hygiene checks pass.
 - **TWAP / Freshness Guards** — Price-quality checks (time-weighted average, maximum quote age, deviation fence) that must pass before any buyback executes; otherwise the controller does nothing (**fail-closed**).
 - **Per-Identity Vote Cap (0.1%)** — The maximum counted voting power per verified human, applied after delegation so no identity can exceed 10 bps of total supply.
-

Part 6 — Deliverables Recap

This appendix bundle gives: (i) an **audit-ready KPI dictionary**; (ii) a **full parameter annex** ready to seed `config@v1`; (iii) standardized **PSIP templates** for C1/C2/C3; (iv) operational **checklists** that gate releases, buybacks, identity, Store hygiene, and the Monthly Digest; (v) concise, versioned **receipt schemas**; and (vi) a **custody registry skeleton** and **DAO-IDEAS pilot map**. Each item cross-references Peace Soldiers Whitepaper & Tokenomics and maps cleanly to code and process so reviewers can verify, not guess.